

Cortical/Functional Areas of the Brain and Lesions

- 3 kinds: motor, sensory, & association.

1. Motor areas

- Found in the frontal lobe
- 4 areas
- a. Primary motor cortex – PMC (4):** In precentral gyrus & anterior part of paracentral lobule.
 - Size of functional area is directly proportional to skilled movement, not to size of muscle.
 - Initiates fine extreme voluntary movements (contralateral).
- b. Premotor cortex (6, 8, 44, 45):** Anterior to precentral gyrus.
 - Controls learned, repetitious, or patterned motor skills
 - Involved in the planning of movements
 - Broca's area (44, 45) – motor area for speech.
 - o Lesion → motor (Broca's) aphasia
 - Frontal eye field area (8) – voluntary eye movement
 - o Fix eye on object persistently
 - Head rotation area (6)
 - Hand skill area (part of 6) - Store motor program for hand skill movement
 - o Lesion → motor apraxia (Absence of skilled hand movement)x
- c. Supplementary motor cortex:** On the medial surface of the premotor cortex.
- d. Prefrontal cortex:** Anterior to premotor cortex.
 - Intellect, Judgment, Prediction, Motivation, & Planning of behavior.

2. Sensory areas

- a. Primary somatosensory cortex (1, 2, 3)**
 - In the postcentral gyrus & posterior part of paracentral lobule.
 - Termination of thalamocortical fibers.
- b. Primary visual cortex (17)**
 - Receives optic radiation from lateral geniculate nucleus (LGN) of the thalamus.
 - Lesion → homonymous hemianopia
- c. Primary auditory cortex (41, 42)**
 - Receives from medial geniculate nucleus (MGN) of the thalamus.
- d. Olfactory cortex-** Piriform lobe & Uncus
- e. Gustatory cortex (43) -** In parietal lobe deep to temporal lobe

3. Association areas

- Interpretation of sensory information.
- a. Visual Association Areas (18, 19) -** integrates & interprets visual information, by relating current impulse to past experiences & knowledge
 - o Lesion → Visual agnosia, (inability to recognize a seen object)
- b. Somatosensory association areas (5, 7, 40)-** understanding (size, texture, and relationship of parts) of the stimulus
- c. Angular Gyrus:** Visual association area (B39) - recognition of visual symbols
 - a. Lesion → Visual aphasia
- d. Auditory Association Area (22) -** comprehension of speech/interpretation of sound info.

- a. Lesion → Auditory (Wernicke's) aphasia (Lack of comprehension of words by vision or hearing)

e. General (Common) Interpretation:

- a. Ill-defined region including parts of the temporal, parietal, and occipital lobes
- b. Found in one hemisphere, usually the left
- c. Integrates incoming signals into a single thought
- d. Involved in processing spatial relationships

f. Visceral Association Area - Located in the cortex of the insula

- a. Involved in conscious perception of visceral sensations

Vasculature of the Brain and Spinal Cord and Lesions

BLOOD SUPPLY TO BRAIN

- 2 major arteries:
- Internal carotid artery: ACA & MCA
- Vertebral artery: basilar, PCA, & anterior spinal artery

Cerebrum: by 3 arteries

Artery	Areas supplied	Occlusion
ACA	- Medial surface of frontal & parietal lobes - Anterior 4/5th of corpus callosum - Ant. portions of basal ganglia & internal capsule	- Paralysis & sensory loss of contralateral foot & leg
MCA	- Most of lateral surface of the hemisphere - Part of internal capsule & basal ganglia.	- Paralysis & sensory loss of contralateral face & arm. - Aphasia (if left)
PCA	- Occipital lobe; lateral aspect of MO - Inferior temporal gyri & uncus; Thalamus - Walls, & choroid plexus of 3rd ventricle	- Contralateral loss of pain & T sensations - Loss of Vision - Ipsilateral deficits of CNs V, VIII, IX, X, & XI - Horner Syndrome; Prosopagnosia

Thalamus

- Branches of **PCA**: -polar & paramedian arteries, inferolateral (thalamogeniculate) arteries, and posterior (medial and lateral) choroidal arteries.

Midbrain

- PCA, sup. cerebellar artery, basilar branches

Medulla Oblongata

- Anterior spinal, PICA, direct branches of vertebral.

Cerebellum

- Superior cerebellar & anterior inferior cerebellar arteries: branches of basilar

- Posterior inferior cerebellar artery (PICA): branch of vertebral
 - o Occlusion: lateral medullary (Wallenberg) Syndrome

BLOOD SUPPLY TO SPINAL CORD

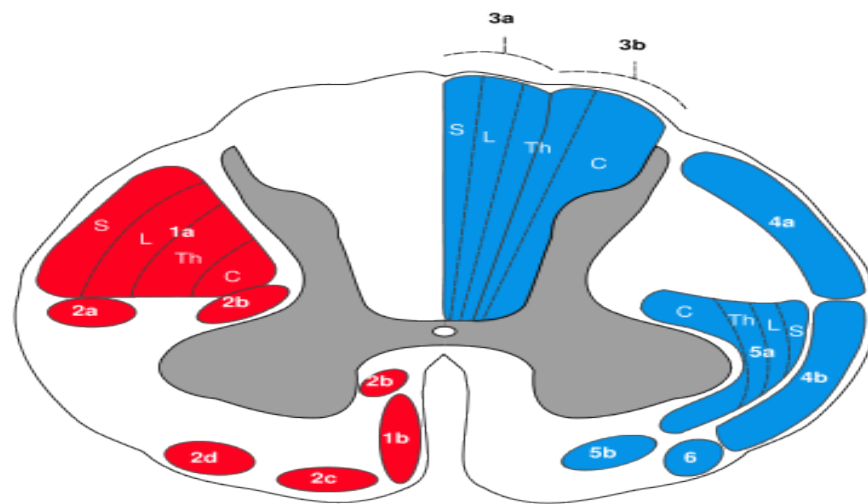
- Anterior spinal arteries (2): from vertebral
- Posterior spinal arteries (2): from vertebral or PICA
- Radicular arteries: from the vertebral, ascending cervical, intercostals & lumbar arteries

White-Matter of the Brain and Spinal Cord

White matter of cerebrum

- Made of myelinated axons
- Gray structures within the white matter → basal ganglia
- Subcortical fibers: 3
 1. Association fibers: within the same hemisphere.
 2. Commissural fibers: connect the two hemispheres.
 3. Projection fibers: connect cerebral hemisphere with other part of CNS.

White matter of spinal cord



Motor and descending (efferent) pathways (left, red)

1. Pyramidal Tracts

- 1a. Lateral corticospinal tract
- 1b. Anterior corticospinal tract

2. Extrapyramidal Tracts

- 2a. Rubrospinal tract
- 2b. Reticulospinal tract
- 2c. Vestibulospinal tract
- 2d. Olivospinal tract

Somatotopy Abbreviations:

S: Sacral, L: Lumbar
Th: Thoracic, C: Cervical

Sensory and ascending (afferent) pathways (right, blue)

3. Dorsal Column Medial Lemniscus System

- 3a. Gracile fasciculus
- 3b. Cuneate fasciculus

4. Spinocerebellar Tracts

- 4a. Posterior spinocerebellar tract
- 4b. Anterior spinocerebellar tract

5. Anterolateral System

- 5a. Lateral spinothalamic tract
- 5b. Anterior spinothalamic tract

- 6. Spino-olivary fibers

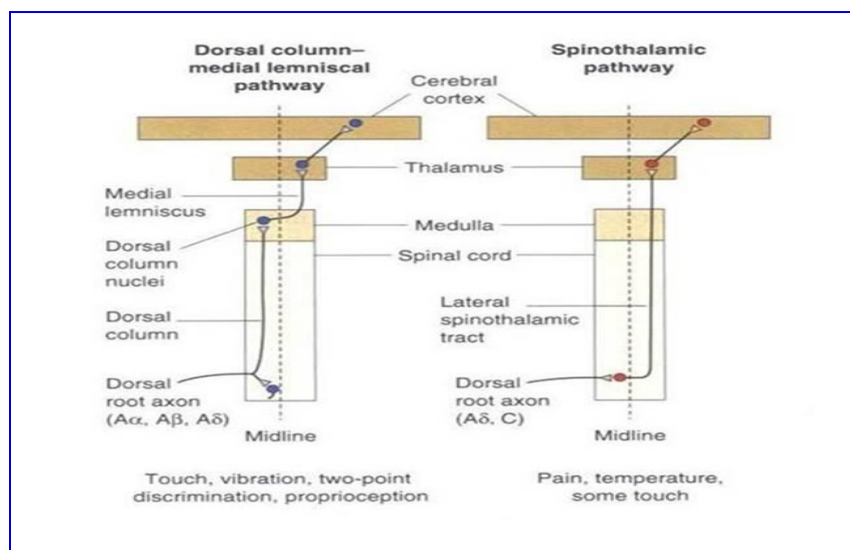
- Anterior, lateral, & posterior funiculus or column.

- Notice the somatotopy of most tracts
 - o Lateral spinothalamic & corticospinal upper body → medially
 - o Fasciculus Gracilis → lower part of the body & cuneatus → upper
 - Used to differentiate if a section of spinal cord is from cervical or lumbar (in both cases we have enlargement but in the lumbar we find only fasciculus gracilis & in cervical both)
 - o Anterior Spinocerebellar → from lower body thru SCP
 - o Posterior Spinocerebellar → from upper body thru ICP
- Fibers run in three directions – ascending (sensory), descending (motor), & intersegmental
- Pathways decussate
- Pathways are paired (one on each side of spinal cord or brain)

Ascending pathways:

For conscious perception:

- Spinothalamic (Anterolateral) system
 - o Lateral: *Pain & temperature*
 - Decussate at level of entry
 - o Anterior: *Light touch & pressure*
 - o Second order fibers: → Traveling in anterolateral parts of spinal cord & in posterolateral parts of brainstem & Terminate in VPL nuclei of thalamus.
- Dorsal Column/Medial Lemniscal system
 - o *Proprioception (position sense), well localized sensation of fine touch, two point discrimination, Vibration sense.*
 - o Fibers enter ipsilateral dorsal funiculus without synapsing until nucleus cuneatus & nucleus gracilis in lower medulla
 - o Second-order fibers then cross the midline & ascend as the medial lemniscus to the VPL nuclei of the thalamus
 - o Third-order thalamocortical fibers terminate in postcentral gyrus.



- Trigeminal pathway
 - o Pain, temperature, proprioception & fine touch sensations from face
 - o The trigeminal ganglion represents the 1st order neuron, except those transmitting proprioception.
 - o 1st order neuron enter lateral aspect of the pons & divide into ascending & descending branches which terminate in the sensory nucleus of CN – V
 - o 2nd order neurons terminate in VPM of thalamus

For unconscious perception:

- Spinocerebellar, Spino-olivary, Spinotectal. Spinoreticular

Descending pathways

Pyramidal system

- ▶ Corticospinal, Corticobulbar tract

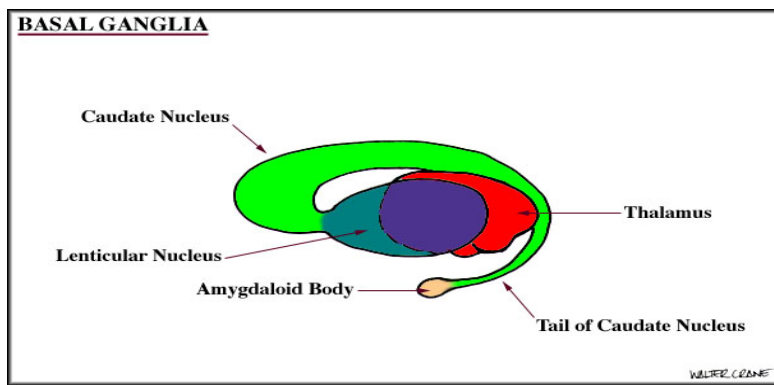
Extra pyramidal system

- ▶ Reticulospinal tract, Rubrospinal tract, Tectospinal tract, Vestibulospinal tract, Olivospinal tract

Gray matter of brain and spinal cord

Gray matter of brain

Cerebrum- basal ganglia



Corpus Striatum: Divided into caudate & lentiform nuclei by anterior limb of internal capsule.

Caudate nucleus: lies lateral to thalamus.

Head of caudate forms the lateral boundary of lateral ventricles

Amygdala is found in the tail of caudate

Lentiform nucleus: is divided into putamen & globus pallidus

Substantia nigra: separates crus cerebri from tegmentum

- Responsible for dopamine production in the brain, thus plays a vital role in reward (pleasurable effects) & addiction

The claustrum: a thin layer of grey matter lying b/n extreme capsule & external capsule

- Function not clearly known

Connections of Corpus Striatum:

- ▶ caudate nucleus & putamen form main sites of receiving input to the basal nuclei
- ▶ Globus pallidus forms the major site of output of the basal nuclei

Gray matter of spinal cord

- ▶ In cross-section → H-shaped:
 - Anterior gray horn → motor, contain large multipolar neurons
 - Posterior gray horn → sensory
 - Lateral gray horn (thoracic & upper lumbar: → T1- L2/3) → SNS
- ▶ Anterior & posterior horns are attached by gray commissure: → central canal
- ▶ Size of gray matter, related to amount of muscle innervated

Diencephalon

- Contains primarily thalamus, hypothalamus, & epithalamus

Thalamus

- ▶ Processes & relays sensory information to cerebral cortex.
- Relay center for sensory systems: except the olfactory.
- also has role in "motor" systems
- Plays an important role in regulating states of sleep & wakefulness.
- ▶ Reciprocal connections with cerebral cortex, forming thalamo-cortico-thalamic circuits.

Hypothalamus

- ▶ Lie in Floor & lateral wall of 3rd Ventricle; & Roof of Interpeduncular Fossa found in all mammalian brains.
- ▶ Links nervous system to endocrine system via pituitary gland (hypophysis).
- ▶ Regulates metabolic processes & activities of ANS.
- ▶ Synthesizes & secretes hypothalamic-releasing hormones, which, in turn, stimulate or inhibit secretion of pituitary hormones.

- ▶ Controls body temperature, hunger, thirst, fatigue, anger, & circadian cycles.

Epithalamus

- Main component: pineal gland (circadian rhythm)

THE OTHER TITLES HAVE NOT BEEN EDITED DUE TO LACK OF VOLUNTEERS.

Histology of the NS Glial cells are 10 times more abundant in the mammalian brain than neurons
 Glial cells of the CNS: Microglial cells, Astrocytes, Oligodendrocytes, Ependymal cells
 Glial cells of the PNS: Schwann cells, Sattelite cells
 Oligodendrocytes: produce the myelin sheath that provides the electrical insulation for neurons in the CNS. Astrocytes
 Astrocytes have supportive roles for neurons and are very important for proper formation of the CNS during embryonic and fetal development
 Some astrocytes develop processes with expanded perivascular feet that cover capillary endothelial cells and contribute to the blood-brain barrier. Other expanded processes form a layer, the superficial glial limiting membrane which lines the pia mater, the innermost meningeal layer at the external surface of the CNS. Furthermore, when the CNS is damaged, astrocytes proliferate to form cellular scar tissue (which often interferes with neuronal regeneration). They regulate constituents of the extracellular environment, absorb local excess of neurotransmitters, and secrete numerous metabolites and factors regulating neuronal activities.
 Ependymal cells • Ependymal cells are low columnar or cuboidal cells that line the ventricles of the brain and central canal of the spinal cord • In some CNS locations, the apical ends of ependymal cells have cilia, which facilitate the movement of cerebrospinal fluid (CSF), or long microvilli, which are likely involved in absorption
 Microglia • microglia are small cells with short irregular processes • analyzing the tissue for damaged cells and invading microorganisms. • They secrete a number of immunoregulatory cytokines and constitute the major mechanism of immune defense in CNS tissues.
MEDICAL APPLICATION: In multiple sclerosis, the myelin sheath is damaged by an autoimmune mechanism with various neurologic consequences. In this disease, microglia phagocytose and degrade myelin debris by receptor-mediated phagocytosis and lysosomal

activity. Satellite cells of the ganglia • small satellite cells form a covering layer over the large neuronal cell bodies in PNS ganglia
 Closely associated with the neurons, Schwann Cells (Neurolemmocytes) • Have trophic interactions with axons and allow for their myelination like the oligodendrocytes of the CNS. • One neurolemmocyte forms myelin around a segment of one axon, in contrast to the ability of oligodendrocytes to branch and sheath parts of more than one axon.
 CNS • It has virtually no connective tissue and is therefore a relatively soft, gel-like organ • white (white matter) and gray (gray matter), differences caused by the differential distribution of myelin
 Gray matter: contains: • abundant neuronal cell bodies, Dendrites, the initial unmyelinated portions of axons, astrocytes, and microglial cells. Types of nerve cells are present in the cerebral cortex: 1) Pyramidal cells- if large betz cells 2) Stellate cells 3) Fusiform cells 4) Horizontal cells of Cajal 5) Cells of Martinotti
 Neuronal connections of the cerebral cortex
 Layer I - Molecular layer (plexiform layer) □ • This is the most superficial layer □ large numbers of synapses
 Layer II- External granular layer □ small pyramidal cells and stellate cells □ The

dendrites of these cells terminate in the molecular layer □ The axons enter deeper layers, where they terminate or pass on to enter the white

matter of the cerebral hemisphere Layer III- External pyramidal layer □ pyramidal cells □ The apical dendrites pass into the molecular layer □ The axons enter the white matter as projection, association, or commissural fibers Layer IV- Internal granular layer □ This layer is composed of closely packed stellate cells □ There is a high concentration of horizontally arranged fibers known collectively as the external band of Baillarger Layer V – Ganglionic layer (internal pyramidal layer) □ very large and medium-size pyramidal cells □ horizontally arranged fibers that form the inner band of Baillarger □ In the motor cortex of the precentral gyrus, the pyramidal cells are very (Betz cells) Layer VI- Multiform layer (layer of polymorphic cells) • Although the majority of the cells are fusiform Variations in Cortical Structure □ Not all areas of the cerebral cortex possess six layers □ Those areas of the cortex in which the basic six layers cannot be recognized are referred to as heterotypical □ Homotypical - possess six layers □ Two heterotypical areas are described: 1. The granular and 2. The agranular type. Granular type, • layers 2 and 4 are well developed, and merge into a single layer of predominantly granular cells • Layers 3 and 5 are poorly developed, • • It is these cells that receive thalamocortical fibers

□ The granular type of cortex Is found in • The postcentral gyrus, superior temporal gyrus, Parts of the hippocampal gyrus Agranular type of cortex • The pyramidal cells in layers 3 and 5 are densely packed and are very large • The agranular type of cortex is found in precentral gyrus and other areas in the frontal lobe Structure of the Cerebellum 1. Outer covering of gray matter called the cortex 2. Inner white matter □ Embedded in the white matter of each hemisphere are three masses of gray matter forming the intracerebellar nuclei Structure of the Cerebellar Cortex □ The cut surface of white matter has a branched appearance, called the arbor vitae Cerebellum □ The gray matter of the cortex throughout its extent has a uniform structure □ It may be divided into three layers: (1) An external layer, the molecular layer; (2) A middle layer, the Purkinje cell layer; and (3) An internal layer, the granular Molecular Layer Contains two types of neurons: 1. The outer stellate cell and 2. The inner basket cell Purkinje Cell Layer □ The Purkinje cells are large Golgi type I neurons

□ They are flask shaped □ Arranged in a single layer □ Dendritic spines form synaptic contacts with the parallel fibers derived from the granule cell axons □ At the base of the Purkinje cell, the axon arises and passes through the granular layer to enter the white matter, terminates by synapsing with cells of one of the intracerebellar nuclei Granular Layer □ The axon of each granule cell passes into the molecular layer, where it bifurcates at a T junction, the branches running parallel to the long axis of the cerebellar folium □ These fibers, known as parallel fibers, run at right angles to the dendritic processes of the Purkinje cells □ Most of the parallel fibers make synaptic contacts with dendrites of the Purkinje cells Dura Mater • dura mater is separated from the periosteum of the vertebrae by the epidural space, which contains a plexus of thin-walled veins and areolar connective tissue. • The dura mater is always separated from the arachnoid by the thin subdural space. • The internal surface of all dura mater, as well as its external surface in the spinal cord, is covered by simple squamous epithelium of mesenchymal origin. Arachnoid • Trabecular spaces are filled with CSF. • In some areas, the arachnoid

perforates the dura mater and protrudes into blood-filled venous sinuses within the dura mater. • Their function is to transport CSF from the subarachnoid space into venous sinuses. Pia Mater • Between the pia mater and the neural elements is a thin limiting layer of astrocytic processes, which adheres firmly to the pia mater. • Blood vessels penetrate the CNS through tunnels covered by pia mater—the perivascular spaces. Blood-Brain Barrier • The main structural component of the BBB is the capillary endothelium, in which the cells are tightly sealed together with well-developed occluding junctions and show little or no transcytosis. • Moreover, the perivascular feet of astrocytes which further regulate passage of

molecules and ions from blood to brain. Choroid Plexus • It is found in the roofs of the third and fourth ventricles and in parts of the walls of the two lateral ventricles, all regions in which the ependymal lining directly contacts the pia mater. • Each villus of the choroid plexus contains a thin layer of well-vascularized pia mater covered by cuboidal ependymal cells. • The main function of the choroid plexus is to remove water from blood and release it as cerebrospinal fluid (CSF).

The eye

- tunica fibrosa, or tunica externa/corneoscleral coat, which consists of the sclera(the white/opaque portion) and the cornea(translucent portion).
- tunica vasculosa also called vascular coat, the middle layer, or uvea, consisting of the choroid, the ciliary body, and the iris
- retina that consists of the pigment epithelium and the neural layer plus the epithelium of the ciliary body and iris(which are extensions of the retinal layers)

The neural retina is continuous with the central nervous system through the optic nerve

Tunica fibrosa

THE SCLERA

- comprises the posterior five sixths of the outer surface of the eyeball
- The extraocular muscles, which move the eyes in their orbits, insert in the sclera.
- Relatively avascular

THE CORNEA

- is a transparent tissue, colorless
- It consists of five layers.

1. An external stratified squamous epithelium,
2. An anterior limiting membrane (Bowman's membrane, the basement membrane of the stratified epithelium),
3. The stroma,

4. A posterior limiting membrane (Descemet's membrane, the basement membrane of the endothelium), and

5. An inner simple squamous endothelium

Corneal epithelium

- stratified squamous epithelium rests on the Bowman membrane.

- Numerous mitotic figures are present in the basal layers, particularly near the periphery of the cornea, reflecting the epithelium's high capacity for cell renewal and repair.

Bowman's membrane

- It does not regenerate. Therefore, if damaged, an opaque scar forms that may impair vision

Stroma

- The thickest layer, the stroma, comprises 90% of the thickness of the cornea

Descemet membrane

- the basal lamina of corneal endothelial cells.

- Unlike Bowman's membrane, Descemet's membrane readily regenerates after injury.

Corneal endothelium

- corneal endothelium (simple squamous epithelium) that is only one cell thick and rests on the Descemet membrane

- provides for metabolic exchange between the cornea and aqueous humor.

The corneoscleral limbus is the transitional zone between the cornea and sclera.

The overlying epithelium at this site thickens from the 5 cell layers of the cornea to the 10 to 12 cell layers of the conjunctiva.

In the stromal layer, endothelium-lined channels called the trabecular meshwork (or spaces of Fontana) merge to form the scleral venous sinus (canal of Schlemm). This sinus encircles the eye.

Changes in the iridocorneal angle may lead to blockage in the drainage of aqueous humor causing glaucoma

Vascular layer

The choroid

- is the highly vascular, pigmented layer of the eye

- The abundant melanocytes give - black color not only to the choroid but also to iris and block light from entering the eye except through the pupil.

LAYERS OF THE CHOROID

1. Suprachoroidal lamina
2. Choriocapillary lamina
3. Bruch's membrane

- The outer part of the choroid bound to the sclera is the suprachoroidal lamina.
- The inner region is richer than the outer layer in microvasculature and is called the choriocapillary lamina.
- Its microvasculature is important for nutrition and normal maintenance of the retina.
- A thin (2–4 m) amorphous hyaline sheet known as Bruch's membrane separates the choriocapillary layer from the retina.

Ciliary body

- It is lined on its inner surface by ciliary epithelium which is two layers of cells:
 1. an outer pigmented columnar epithelium rich in melanin/ is contiguous with the sensory layer of the retina.
 2. an inner nonpigmented simple columnar epithelium/ Correspond to the anterior projection of the pigment layer of the retina.
- Ciliary muscle contraction in response to parasympathetic permits the lens to become more convex, allowing the eye to focus on nearby objects (accommodation). With advancing age, the lens loses its elasticity, thereby gradually losing the ability to accommodate.
- This fluid has an inorganic ion composition similar to that of plasma but contains less than 0.1% protein (plasma has about 7% protein).

Iris

- is the most anterior extension of the uvea,
- THE ANTERIOR SURFACE of the iris, exposed to the anterior chamber, is not covered by epithelium, but consists of an irregular, discontinuous layer of fibroblasts and melanocytes.
- The iris is covered on its inner surface by a two-layered epithelium:
 1. pigmented cells are filled with melanin granules that block light from entering the interior of the eye except via the pupil.
 2. myoepithelial cells - Radially extended processes from these myoepithelial cells make up the very thin dilator pupillae muscle along the posterior side of the iris.
- Eye color is blue if only a few melanocytes are present. Increasing amounts of pigment impart darker colors to the eye.

- Dilator pupillae muscle. It contracts upon stimulation by sympathetic nerve fibers, dilating the pupil
- Sphincter pupillae muscle. It contracts upon stimulation by parasympathetic nerve fibers of the oculomotor nerve (cranial nerve III), constricting the pupil.

Lens

- It is highly elastic, a feature that is lost with age as lens tissue hardens. The lens has three principal components
- composed of the lens capsule, subcapsular epithelium, and lens fibers
- Lens fibers represent highly differentiated, elongated cells. Developing from stem cells in the lens epithelium, the differentiating lens fibers eventually lose their nuclei and other organelles, fill the cytoplasm with a group of proteins called crystallins, and become very long.

Cataract

1. A cataract is an opacity of the lens resulting from the accumulation of pigment or other substance in the lens fibers.

Presbyopia : is the inability of the eye to focus on close objects (accommodation

Retina

- The retina, the inner layer of the eye, is derived from the embryonic optic cup. Like the optic cup, the retina consists of two major layers.

- 1) The inner one, the neural retina, contains the neurons and photoreceptors
- 2) The outer pigmented layer is an epithelium resting on Bruch's membrane just inside the choroid.

In the neural retina, two regions or portions that differ in function are recognized:

- 1) The nonphotosensitive region (nonvisual part), located anterior to the ora serrata
- 2) The photosensitive region (optic part) lines the inner surface of the eye posterior to the ora serrata

- OPTIC DISC - The site where the optic nerve joins the retina is called the optic disc or optic papilla. Because the optic disc is devoid of photoreceptor cells, it is a blind spot in the visual field.

- THE FOVEA CENTRALIS is a shallow depression located about 2.5 mm lateral to the optic disc. It is the area of greatest visual acuity. It that contains only cones; this avascular region, whose central region, the macula, exhibits the greatest visual acuity

The retina has 10 layers

The retinal pigment epithelium

- is a layer of columnar cells firmly attached to the Bruch membrane.
- Retinal pigment epithelial cells esterify vitamin A (used in the formation of visual pigment by rods and cones).
- They synthesize melanin, which absorbs light after the rods and cones have been stimulated

The ganglion cell layer:

- Ganglion cells are activated by hyperpolarization of rods and cones and generate an action potential

The optic nerve fiber layer

- consists primarily of the unmyelinated axons of ganglion cells, which form the fibers of the optic nerve. As each fiber pierces the sclera, it acquires a myelin sheath.

Detachment of the retina

1. Retinal detachment occurs when the neural and pigmented layers separate from each other, for example, as a result from a sudden hard jolt.

The ear

External ear

1. The auricle (pinna) is composed of irregular plates of elastic cartilage covered by thin skin.
2. The external auditory meatus is lined by skin containing hair follicles, sebaceous glands, and ceruminous glands
 - A canal extending from the lateral surface of the head to the ear drum .
 - It is lined with stratified squamous epithelium continuous with the skin of the auricle and
 - near its opening hair follicles, sebaceous glands, and modified apocrine sweat glands called ceruminous glands are found in the submucosa.
 - Cerumen is the oily or waxy, yellowish material resulting from secretions of the sebaceous and ceruminous glands.

3. Tympanic membrane (eardrum)

- The tympanic membrane is covered by skin on its external surface and by a simple cuboidal epithelium on its inner surface continuous with the lining of the tympanic cavity in the middle ear.

Middle ear (tympanic cavity)

1. Air filled cavity

2. Two tiny skeletal muscles (tensor tympani and stapedius)- insert into the malleus and stapes, restricting movement of the ossicles and helping to protect the internal ear from extremely loud noises.

3. The middle ear is connected to the pharynx via the auditory tube(eustachian tube).

Conductive hearing loss

1. Conductive hearing loss results from a defect in the conduction of sound waves in the external or middle ear.

2. It may be caused by otitis media,a common inflammation of the middle ear; obstructionby a foreign body; or otosclerosis of the middle ear.

Internal Ear

☐ Lies within the thick, protective walls of the petrous part of the temporal bone.

☐ The internal ear consists of two main divisions:

1. the bony labyrinth and

2. the membranous labyrinth

The bony labyrinth,composed of the semicircular canals, vestibule, and cochlea are filled with perilymph, and house the membranous labyrinth filled with endolymph.

The perilymph is continuous with the cerebrospinal fluid that fills the subarachnoid space.

The membranous labyrinth is a continuous series of membrane-walled sacs and ducts that fit loosely within the bony labyrinth and more or less follow its contours.

The wall of the membranous labyrinth—its “membrane”—is a thin layer of connective tissue lined by a simple squamous epithelium.

Parts of this epithelium are thickened and contain the receptors for equilibrium and hearing.

• The main parts of the membranous labyrinth are

1. The semicircular ducts, one inside each semicircular canal;

2. The utricle and saccule, both in the vestibule; and

3. The cochlear duct in the cochlea.

Cochlea

- It is subdivided into three spaces: the scala vestibuli and scala tympani, which are both filled with perilymph, and the scala media, or cochlear duct, which is filled with endolymph.

- membranous labyrinth within the cochlea is called the cochlear duct. This is the location of the sensory receptors for hearing.

- The “roof” of the cochlear duct, separating it from the scala vestibuli, is the vestibular membrane.
- The floor of the cochlear duct consists of the osseous spiral lamina plus an attached sheet of fibers called the basilar membrane.
- The basilar membrane supports the spiral organ (of Corti), the receptor epithelium for hearing.
- This tall epithelium consists of columnar supporting cells and one row of inner and three rows of outer hair cells, which are the receptor cells.
- At the cell’s apex, the tips of the hairs, the stereocilia, are embedded in a gel-like tectorial membrane (“roofing membrane”);
- At their base the hair cells synapse with sensory fibers of the cochlear nerve.
- These nerve fibers belong to bipolar neurons, whose cell bodies occupy a spiral ganglion in the osseous spiral lamina and modiolus

Hair cells of the organ of Corti

- possess many long, stiff stereocilia (arranged in a W formation) on their free surfaces, as well as a basal body (but no kinocilium).

Stria vascularis

- is a vascularized pseudostratified epithelium that lines the lateral aspect of the cochlear duct.
- may secrete endolymph.

The Vestibule

- Suspended within its perilymph are the two egg-shaped parts of the membranous labyrinth, the utricle (“leather bag”) and the saccule (“little sac”).
- The utricle is continuous with the semicircular ducts; the saccule, with the cochlear duct.

The utricle and saccule

- each house a spot of sensory epithelium called a macula. macula of the saccule contain receptor cells that monitor the position of the head when the head is held still. This aspect of the sense of balance is called static equilibrium.
- These receptor cells also monitor straight-line changes in the speed and the direction of head movements—that is, linear acceleration—but not rotational movements of the head.
- Each macula is a patch of epithelium containing columnar supporting cells and scattered receptors called hair cells.
- The tips of these stiff hairs are embedded in an overlying otolithic membrane, which is actually a jellylike disc that contains heavy crystals of calcium carbonate called otoliths (“ear stones”).

Semicircular canal

- The ampullae are dilated regions of the semicircular ducts near their junctions with the utricle.
- Each membranous ampulla houses a small crest called a crista ampullaris, or “crest of the ampulla”
- Cristae ampullares are specialized sensory regions within the ampullae of the semicircular ducts. They are similar to maculae but have a thicker, conical glycoprotein layer (cupula), which does not contain otoliths.
- Each crista has an epithelium on its top that, like the maculae, contains supporting cells and receptor hair cells.
- The “hairs” of these hair cells project into a tall, jellylike mass that resembles a pointed cap, the cupula
- Spinning the body can produce vertigo due to overstimulation of the cristae ampullares of the semicircular ducts.
- Overstimulation of the maculae of the utricle caused by repetitive changes in linear acceleration and directional changes can normally lead to motion sickness (seasickness)

Ménière disease

1. Ménière disease is an inner ear disorder causing hearing loss, vertigo, nausea, tinnitus, and vomiting. It is related to excess fluid in the endolymphatic duct.

Development of the nervous system

The nervous system develops from the neural plate, a thickened area of embryonic ectoderm.

- Formation of the neural folds, neural tube, and neural crest from the neural plate

A. CNS develops from a dorsal thickening of ectoderm-the neural plate, which appears around middle of the third week, neural plate into neural tube to give the brain and spinal cord

Neurulation-formation of the neural plate and neural tube. It begins during stage 10 of development (22-23 days)

The notochord induces the overlying ectoderm to differentiate into neuroectoderm and form the neural plate. The notochord forms the nucleus pulposus of the intervertebral disk in the adult.

1. The anterior neuropore closes during week 4 (day 25) and becomes the lamina terminalis. Failure of the anterior neuropore to close results in upper neural tube defects (NTDs; e.g., anencephaly).
2. The posterior neuropore closes during week 4 (day 27). Failure of the posterior neuropore to close results in lower NTDs (e.g., spina bifida with myeloschisis).

cranial two thirds of the neural plate & tube, cranial to 4th pair of somites, represent the future brain, and caudal one third of the neural plate & tube represents the future spinal cord.

The neural canal forms the ventricular system of the brain and the central canal of the spinal cord.

Nonclosure of the Neural Tube:

- The current hypothesis is that there are multiple, possibly five, closure sites involved in formation of the neural tube.
- Failure of closure of site 1 results in spina bifida cystica;
- meroencephaly (anencephaly) results from failure of closure of site 2;
- craniorachischisis results from failure of sites 2, 4, and 1 to close;

Fusion of the neural folds in the cranial region and closure of the rostral neuropore form three primary brain vesicles from which the brain develops .

A. The three primary brain vesicles and two associated flexures develop during week 4.

1. Primary brain vesicles

- a. Prosencephalon (forebrain) is associated with the appearance of the optic vesicles and gives rise to the telencephalon and diencephalon.
- b. Mesencephalon (midbrain) remains as the mesencephalon.
- c. Rhombencephalon (hindbrain) gives rise to the metencephalon and myelencephalon.

2. Flexures

During the fourth week, the embryonic brain grows rapidly and bends ventrally with the head ion of hindbrain & spinal cord.

The pontine flexure, located in the future pontine region, divides the hindbrain into caudal (myelencephalon) and rostral (metencephalon) parts.

- The myelencephalon becomes the medulla oblongata & the metencephalon becomes the pons and cerebellum.
- The cavity of the hindbrain becomes the fourth ventricle and the central canal in the medulla.

B. The five secondary brain vesicles become visible in week 6 of development and form various adult derivatives of the brain.

1. Telencephalon gives rise to the cerebral hemispheres, caudate, and putamen.

2. Diencephalon gives rise to the epithalamus, subthalamus, thalamus, hypothalamus, mammillary bodies, neurohypophysis, pineal gland, globus pallidus, retina, iris, ciliary body, optic nerve (CN II), optic chiasm, and optic tract.

3. Mesencephalon gives rise to the midbrain.
4. Metencephalon gives rise to the pons and cerebellum.
5. Myelencephalon gives rise to the medulla.

medulla

Basal plates form motor CNs: V, VIII, IX, X & gracil, cuneate & olivary nuclei

The roof plate forms the roof of the fourth ventricle. The roof plate is called the tela choroidea, which is a monolayer of ependymal cells covered with pia mater. The tela choroidea is invaginated by pial blood vessels to form the choroid plexus of the fourth ventricle.

Resorption of roof plate result in foramen of luschka & magendie

The remanant of roof plate form median sulcus

Metencephalon:

- The walls of the metencephalon form the pons and cerebellum,
- and the cavity of the metencephalon forms the superior part of the fourth ventricle

Pons

Arise from anterior aspect of metencephalon

- Basal plates form motor nuclei of: CNs V, VI, & VII
- Alar plates form sensory nuclei of CNs: V, VII, VIII & pontine nuclei

Mid brain

Basal plates – form nuclei of CN III, IV, red nuclei, substantia nigra & reticular formation

- Marginal zone - forms Cerebral Peduncle
- Alar plates- with roof plate form the tectum & sensory neurons of colliculi

Mid brain undergoes less change than any other part of the developing brain • The neural canal narrows and becomes the cerebral aqueduct , a channel that connects the third and fourth ventricles

Forebrain:

The rostral or anterior part of the forebrain, including the primordia of the cerebral hemispheres, is the telencephalon,

- and the caudal or posterior part of the forebrain is the diencephalon.
- The cavities of the telencephalon and diencephalon contribute to the formation of the third

The neural tube caudal to the fourth pair of somites develops into the spinal cord

- The lateral walls of the neural tube thicken:

size of the neural canal (central canal) of the spinal cord at 9 to 10 weeks

- Initially, the wall of the neural tube is composed of a thick, pseudostratified, columnar neuroepithelium .
- These neuroepithelial cells give rise to all neurons and macroglial cells (macroglia) in the spinal cord
- Macroglial cells are the larger members of the neuroglial family of cells, which includes astrocytes & oligodendrocytes.

The primordial supporting cells of the central nervous system-glioblasts (spongioblasts)-differentiate from neuroepithelial cells,

- Some glioblasts become astroblasts and later astrocytes, whereas others become oligodendroblasts & eventually oligodendrocytes

l cells, lining the central canal of the spinal cord.

Layers of the neural tube

- Ventricular Zone
- Intermediate Zone :
- Marginal Zone

Development of the spinal cord

The spinal cord develops from the neural tube caudal to the fourth pair of somites.

A. Alar (sensory) plate

The alar plate gives rise to sensory neuroblasts of the dorsal horn (general somatic afferent [GSA] and general visceral afferent [GVA] cell regions).

B. Basal (motor) plate

The basal plate gives rise to motor neuroblasts of the ventral and lateral horns (general somatic efferent [GSE] and general visceral efferent [GVE] cell regions).

C. Sulcus limitans (SL)

1. The SL is a longitudinal groove in the lateral wall of the neural tube that appears during week 4 of development and separates the alar and basal plates.

Neural crest cells

all sensory and autonomic ganglia, • Schwann cells, • meninges of the brain and spinal cord, • adrenal medulla, • pigment cells of the skin • and connective tissues of the head and neck, including odontoblasts for dentin of the teeth

Nerves of Pharyngeal Arches: CNs V, VII, IX, & X:

- supply the embryonic pharyngeal arches; thus, the structures that develop from these arches are innervated by these CNs
- Their ganglia develop from neural crest cells
- trigeminal nerve (CN V) is the nerve of the first pharyngeal arch, but it has an ophthalmic division that is not a pharyngeal arch component.
- facial nerve (CN VII) is the nerve of the second pharyngeal arch.
- glossopharyngeal nerve (CN IX) is the nerve of the third pharyngeal arch.
- vagus nerve (CN X) is formed by fusion of nerves of the 4th & 6th pharyngeal arches
- nerve of the 4th pharyngeal arch becomes the superior laryngeal nerve, which supplies cricothyroid muscle & constrictor muscles of the pharynx.
- nerve of the sixth pharyngeal arch becomes the recurrent laryngeal nerve, which supplies various laryngeal muscles.

Brain malformation

Cranium Bifidum

- Defects in the formation of the cranium (cranium bifidum) are often associated with congenital anomalies of the brain and/or meninges.

Microcephaly:

Exencephaly

- is a severe anomaly of the brain that results from failure of the rostral neuropore to close during the fourth week.
- Most of the embryo's brain is exposed or extruding from the cranium-

Meroencephaly (anencephaly):

- a rudimentary brainstem and functioning neural tissue are always present in living infants.

Hydrocephalus:

- results from impaired circulation and absorption of CSF and, in rare cases, from increased production of CSF by a choroid plexus adenoma.

Neural Tube Defects: Spinal Cord Anomalies

1. Spina Bifida Occulta:

- This defect in the vertebral arch is the result of failure of the embryonic halves of the arch to grow normally and fuse in the median plane

2. Spina Bifida Cystica

- Severe types of spina bifida, involving protrusion of the spinal cord and/or meninges through the defects in the vertebral arches,

Spina bifida cystica is strongly suspected in utero when there is a high level of alpha fetoprotein (AFP) in the amniotic fluid .

Meningomyelocele:

- is a severe type of spinal bifida cystica associated with a marked neurologic deficit inferior to the level of the protruding sac.
- Meningomyeloceles may be covered by skin or a thin, easily ruptured membrane
- Spina bifida with meningocele is a more common and a much more severe anomaly than spina bifida with meningocele.
- These NTDs may occur anywhere along the vertebral column, but they are most common in the lumbar and sacral region.

Myeloschisis :

- This is the most severe type of spina bifida
- In these cases, the spinal cord in the affected area is open because the neural folds failed to fuse
- the caudal neuropore fails to close at the end of the fourth week.
- As a result, the spinal cord is represented by a flattened mass of nervous tissue.

Physiology of the Hypothalamus

- The hypothalamus is the most inferior portion of the diencephalon.
- Located below the thalamus, it forms the floor and part of the lateral walls of the third ventricle.
- Although the hypothalamus occupies less than 1% of total brain volume, it is the center for homeostasis and contains centers for various behavioral drives, such as hunger and thirst.
- Contains neural centers for hunger and thirst and for the regulation of body temperature and hormone secretion from the pituitary gland.
- In addition, centers in the hypothalamus contribute to the regulation of sleep, wakefulness, sexual arousal and performance, and such emotions as anger, fear, pain, and pleasure.
- In its regulation of emotion, the hypothalamus works together with the limbic system.
- Regulation of Autonomic System

- o Experimental stimulation of different areas of the hypothalamus can evoke the autonomic responses characteristic of aggression, sexual behavior, hunger, or satiety.

- ☐ Lateral hypothalamus: Eating and becoming obese.

- ☐ Medial hypothalamus: Inhibits eating.

- o The hypothalamus is also where the body's "thermostat" is located.

- ☐ Preoptic-anterior hypothalamus.

- ☐ Serve to correct the temperature deviations in a negative feedback fashion.

- o The coordination of sympathetic and parasympathetic reflexes is thus integrated with the control of somatic and endocrine responses by the hypothalamus.

- Regulation of the Pituitary Gland

- o The pituitary gland is located immediately inferior to the hypothalamus.

- o Neurons within the supraoptic and paraventricular nuclei of the hypothalamus produce two hormones— antidiuretic hormone (ADH), which is also known as vasopressin, and oxytocin.

- ☐ Oxytocin stimulates contractions of the uterus during labor.

- ☐ ADH stimulates the kidneys to reabsorb water and thus to excrete a smaller volume of urine.

- o Neurons in the hypothalamus also produce hormones known as releasing hormones and inhibiting hormones.

- Regulation of Circadian Rhythms

- o Electrical activity that oscillates automatically in a pattern that repeats about every twenty-four hours.

- ☐ Suprachiasmatic nuclei (SCN).

- ☐ The SCN is believed to be the major brain region involved in regulating the body's circadian rhythms.

- o However, for these neuron clocks to function properly, their activities must be entrained (synchronized) to each other and to the day/night cycles.

- o The daily cycles of light and darkness influence the SCN by way of tracts from the retina (the neural layer of the eyes) to the hypothalamus.

- o The SCN influences the circadian rhythms of the body partly through neural connections to other brain regions, and through its regulation of the pituitary gland, which secretes hormones that stimulate some other endocrine glands. Also, the SCN controls the secretion of the hormone melatonin from the pineal gland.

- Cardiovascular Regulation

- o Stimulation of different areas throughout the hypothalamus can cause many neurogenic effects on the cardiovascular system, including changes in arterial pressure and heart rate.

- o In general, stimulation in the posterior and lateral hypothalamus increases the arterial pressure and heart rate, whereas stimulation in the preoptic area often has opposite effects, causing a decrease in both heart rate and arterial pressure.

- Body Temperature Regulation

- o The anterior portion of the hypothalamus, especially the preoptic area, is concerned with regulation of body temperature.

- o An increase in temperature of the blood flowing through this area increases activity of temperature-sensitive neurons, whereas a decrease in temperature decreases their activity.

- Body Water Regulation

- o Creates sensation of thirst.

- o Control excretion of water in the urine.

- o An area called the thirst center is located in the lateral hypothalamus.

- ☐ When the fluid electrolytes in either this center or closely allied areas become too concentrated, the animal develops an intense desire to drink water.

- o Control of renal excretion of water is vested mainly in the supraoptic nuclei which secrete ADH to act on the kidneys.

- Gastrointestinal and Feeding Regulation

- o One area associated with hunger is the lateral hypothalamic area.

- ☐ Damage to this area on both sides of the hypothalamus causes the animal to lose desire for food, sometimes causing lethal starvation.

- o A center that opposes the desire for food, called the satiety center, is located in the ventromedial nuclei.

- ☐ If this area is destroyed bilaterally, the animal cannot be satiated; instead, its hypothalamic hunger centers become overactive, so it has a voracious appetite, resulting eventually in tremendous obesity.

- Emotion and other behaviors

- o Interacts with limbic system to influence behavior and emotions.

- o Stimulation of visceral responses during emotional situations.

o A major part of the limbic system is the hypothalamus, with its related structures. It sends signals in three directions.

- ☐ Backward and downward to the brain stem.
- ☐ Upward toward many higher areas of the diencephalon and cerebrum.
- ☐ Into the hypothalamic infundibulum.

Higher brain function

Cerebrum: Encode different behaviours: functions of cerebrum like Sleep and wakeful, consciousness, personalities, reasoning, judgment, fine movement, language, learning and memory. Fine and extreme movement is by pyramidal tract

Right hemisphere controls left side of body and numerical skills reasoning. Left hemisphere controls musical and artistic, space and pattern perception imagination spatial memory (navigation).

Frontal lobe function: Personality, behavior, Motor function, Judgments/problem solving
Micturation Speech Concentration Reasoning

Parietal lobe is used in perception of our environment and to know the position of our body parts. Neglecting syndrome: Failure to respond to the novel stimuli presented to the opposite side of the lesion, Caused by parietal lobe lesions, The eye is open, detect light, visual cortex processes the image, but the parietal lobe not respond, Lesion to the right parietal lobe (stroke, tumor), neglect the left side and vice versa. If he draws a flower he can only draw half.

Sensory homunculus: body parts are represented upside down in the primary sensory cortex and The size of the representation is not dependent on the size of the body rather its function The more sensitive the area of the body the more area its mapped

Neuronal function in each layer of brain layer 1: receive signals from the lower brain areas layer 2 and 3 send axons to opposite hemisphere through corpus callosum layer 4 receive signals from thalamus and is most developed in primary sensory cortex. Layer 5 send signals to brain stem basal nuclei and spinal cord. Layer 6 gives signals to thalamus.

Electroencephalogram (EEG): A record of brain electrical activity from the surface of brain (scalp) Multiple electrodes are placed on the scalp, Electrodes are connected with the machine, Changes on the EEG indicates abnormalities: Epilepsy, brain death, coma, tumours, stroke. It measures level of consciousness. In case of ECG waves segments and intervals but in EEG only brain waves. So EEG measures brain waves and The signals are the sum of postsynaptic potentials, Have frequencies inversely proportional with the amplitude, EEG with large amplitude and low frequency shows synchronized (more neurons fire at the same time) wave pattern, The reverse is desynchronized. There are 4 types of brain waves. ALPHA: Awake, non-focused, relaxed, drowsy, Synchronized rhythm and low level of environmental stimulation BETA: Awake, alert, focused attention, problem solving, dream/REM sleep, high level of environmental stimulation (eyes open) & desynchronised rhythm THETA: Visual

imagery, hallucination, light sleep and in children DELTA: Deep and restful sleep, dream states, infancy and in organic brain and also anesthesia Deep or slow-wave sleep or non-REM (NREM) sleep, in which the brain waves are strong and of low frequency,(delta waves)U don't remember the dreams in deep sleep e.g night mares Although slow-wave sleep is frequently called "dreamless sleep," dreams and sometimes even nightmares do occur during slow-wave sleep. The difference between the dreams that occur in slow-wave sleep and those that occur in REM sleep is that those of REM sleep are associated with more bodily muscle activity. Also hormones are release in NREM, NREM sleep takes longer time (75%).

Reticular formation system and consciousness

Diffuse groups of neurons in brainstem Has Ascending Reticular Activating System (ARAS), In the excitatory areas ,Increase the excitability of the cortical neurons and controls level of consciousness,Release Ach into hypothalamus to make it release glutamate to activate cortex. ,Activated by Visual , auditory, pain, drugs and other information.,Anesthetic drugs bring unconsciousness as depressing RAS,Its inhibition (drugs/NTs) causes sleep, Help for consciousness, attention and memory RF HAS EXCITATORY PART THAT RELEASE ACH AND INHIBITORY PART THAT RELEASE GABA.

SLEEP PHYSIOLOGY, HAS 2 PHASES, REM and NREM. EEG changes during these stages

- Sleep theories: Passive Inactivation of the ARAS due to fatigue.

Active: Release of NTS A) Serotonin (5-HT) from raphe magnus to brainstem Reticular formation, Thalamus, Hypothalamus, Most area of the Limbic system, Neocortex of cerebrum. 5-HT also released down into the dorsal horn s.cord that inhibits upcoming signal (pain and other sensation) B) TRACTUS SOLITARIUS ACTIVATION LEADS TO BLOCKAGE OF SIGNALS FROM GLOSSOPHARYNGEAL AND VAGUS NERVE. C) Gaba from ventrolateral preoptic nucleus blocks LH(lateral hypothalamus) (that release orexin wch is awake stimulant) supra chiasmatic nucleus (activates LH) and brain stem (all nuclei here are wake inducing except raphe magnus wch is sleep inducing) :Serotonin . Melatonin , GABA - induce sleep but Ach , glutamate , NE, DA - activate – awake

Limbic system Interconnected group of nuclei surrounding cerebrum, Composed of majorly diencephalons,Regulates emotion, learning, memory, and other behaviors,Hypothalamus is the major components and controls:Behaviors,Vegetative and endocrine body functions Opioids and endorphins are its NT

Communicated with other brain areas Cerebral cortex and Brainstem thus emotion affects thinking and vice versa and emotion affects heart rate and breathing and vice versa. Centers where addictive drugs act AKA Called emotion brain. Hippocampus □ Memory processing and the Amygdala □ Aggression (fight) and fear (flight)

Reward and punishment center in the limbic system

A) Reward center: Lateral and medial nuclei of hypothalamus, Stimulation causes release of DA into NAc(nucleus accumbens) for pleasant sensation For Learning via repetition. If excess NAc stimulation lead to euphoria and addiction.

B) Punishment center: periventricular part of hypothalamus Fear, unpleasant , rage, responses it is how we Learn not to do

FUNCTIONS OF THE LIMBIC SYSTEM: fear and aggression, memory, temperature regulation, smell and navigation e.t.c

Most emotion is controlled by amygdala and it controls it through d/t mechanisms and pathways the major pathway being through hypothalamus for sympathetic control this is called major pathway.

Klüver-Bucy syndrome: Bilateral amygdala ablation, change in behaviors characterized by No fear response, Forget the consequence of the threat Try to put things to mouth and take, Strong sex desire.

Papez's circuit: emotion The interconnected structures in the limbic systems that control emotions.

Language:

Temporal lobe, Frontal lobe, Anterolateral occipital lobe, Angular gyrus: give visually perceived words

Language interpreting Enable us to understand what we see and hear (comprehension) its abnormality : Sensory/ Wernicke's aphasia. Info from somatic sensory area, and also from vision goes to Wernicke's area for understanding it.

Broca's area: In the premotor area For sound creation Its defect unable to sound formation, Motor/Broca's aphasia

Angular gyrus: Its abnormality : subject see the words but not understand (dyslexia/word blindness)

Medial occipital & medioventral temporal lobes (face area) Its abnormality: Prosopagnosia / (face blindness)

Left hemianopsia (left side vision loss when there is right occipital lobe lesion)

Amnesic aphasia: inability to find words to name objects or not to name written objects

Spatial agnosia - Impairment of spatial orientation Topographical disorientation Caused by damage to the parietal and temporal lobes

Learning and memory: Learning: process of experience-dependent acquiring of new info/skill

Memory: Process of recalling and retrieving The mechanism is synaptic plasticity, result of synaptic alterations Activities based changes in synaptic connectivity Short/ long term Internal or external (environmental factors) inputs Short term changes: pre- existing proteins activation

For short term memory Amount of neurotransmitter released at presynaptic nerve, Ionic concentration and Pre existing postsynaptic cell receptor activities

Rat in complex environment: Shows increase in number of neuron synapses by about 25%

Long term changes in memory: New dendrites and dendrite spines formed New proteins (enzymes in the postsynaptic cells) synthesized Protein synthesis inhibitors Deficits in learning and memory New synaptic contacts (Axonal sprouting) Changes on the length of the dendrite spines. it is Glutamate and GABA NTs based

5-10 mints rehearsal is needed for consolidation.

- Learning: none associative.
- Habituation: Learning to ignore stimulus that lacks meaning

Sensitization

- Learning to intensify response to stimuli

Associative: classical conditioning: Pair an unconditional stimulus (UC) with a conditional stimulus (CS) to get a conditioned response (CR)

Instrumental Conditioning

Learn to associate a response with a meaningful stimulus

Retrograde: Loss of memories that were acquired before damaging

Anterograde amnesia: Unable to form new memory after damaging Lesion on the Hippocampus.

PHYSIOLOGY OF SPECIAL SENSES

Vision (Visual System)

Has photo receptors cons and rods wch are also called transducers. Max visual acuity occurs at fovea b/c there are lots of cons there (in central part of retina) in periphery there are more rods.

The Lens

Important for visual refraction (b/c it's transparent); 1/3rd of the refractory power of the eye is attributed to the lens wch has refractive power of 20D and can reach to 34D after accommodation.

The Cornea

- Like the lens, the cornea also has a refractive power. But is much larger (53D). This is simply because the cornea is found more anterior to the object than the lens. It is avascular so transparent.

Aqueous Humor

- A Fluid found between the cornea and the lens (in front of the lens). Ciliary body plays an important role for the formation of aqueous humor. The amount of aqueous humor produced by an average adult per day is 5-6ml. The aqueous has the following important functions:
 - ☐ supplies nutrients to the lens and cornea,
 - ☐ removes wastes
 - ☐ maintains IOP within normal ranges (12-20mmHg) ...otherwise if >20mmHg glaucoma. IOP gives the eye the shape it has. In some cases like infection, the drainage of this fluid is blocked and iop increase. If iop is above 35mmhg it leads to complete blindness.
- There is a certain range of wavelength that a human eye can detect called visible light/spectrum (defined as the wavelength which can stimulate the receptors in the eye). It is the cons which detect color not the rods. This is because the cons have 3 pigments that correspond to the 3 primary colors of the eye.
- The refractive power depends on (1) focal length/distance and (2) thickness (shape) of the lens. The focal distance is the distance between the lens and the focal point (= is specific area on retina where the image is exactly created).
- Refractive power = $1/\text{Focal Distance}$. Hence, if the focal distance is large, the refractive power is low and vidr versa
- When the lens becomes thick (shorter or rounder), the focal distance decreases. Small focal distance means high refractive power. Conversely, when the lens becomes thin (longer), the focal distance increases and hence the refractive power decreases. The shape (thickness) of the lens is controlled by the parasympathetic nervous system (PNS). The PNS thickens the lens hence increases the refractive power.
- Convergence and Divergence property of the lens: these refer to convexity and concavity of the lens respectively. concave lens (-ve value) convex (+ve value). Reduced Eye: the sum of all refractive powers of the eye (that of the anterior, posterior lens & cornea). The D of reduced eye is +59.
- Accommodation: is defined as the ability of the lens to change shape to focus on an object. To focus on a nearer object, the lens should be rounder or shorter (i.e. thicker). To focus on a further object, the lens should be thinner. This is all because of focal length. When a near object is seen, the focal length is large pushing the focal point behind the retina so the thickening of the lens avoids the image from being created at the back of the retina. Likewise, when a far object is seen, the lens should thin out (be flat) to decrease the focal length and avoid the image

from falling short in front of the retina. BASICALLY IT IS TO MAKE THE IMAGE FALL ON RETINA.

- Neuronal mechanism of accommodation: The PNS innervates ciliary muscle of the eye. The lens at both upper and lower ends is attached to the ciliary muscles by fibers called zonular fibers. Up on a parasympathetic innervation, the ciliary muscles contract. As the ciliary muscles contract, the zonular fibers relax (get released...initially these fibers were tightly holding the lenses). This releases the lens from tight attachment/stretching by the zonular fibers. Hence, contraction of the ciliary muscles (up on PNS stimulation) makes the lens thick (round) enabling for a near vision. In the absence of PNS innervation, the ciliary muscles stay relaxed and the lens stay flat (thin) as well enabling for a clear far vision. Parasympathetic postganglionic fibers which innervate the ciliary muscles emerge from ciliary ganglion and release muscarine as a neurotransmitter. The receptors on the ciliary muscle are muscarinic receptors not nicotinic. SNS has no effect on the ciliary muscle.
- Pupillary Light Reflex: When a light of significant intensity is shined into the eyes, the pupils constrict (become narrow). Pupillary constriction is called MIOSIS and is due to PNS (parasympathetic) stimulation of the circular muscle (pupillary sphincter muscle). Pupillary dilation (widening) is called MYDRIASIS and is caused by SNS stimulation of the radial muscle of iris (pupillary dilator muscle). The postganglionic sympathetic fibers emerge from superior cervical sympathetic ganglion to innervate the radial muscle of the iris which causes pupillary dilation. So in summary, the PNS prevents excess light from entering and damaging the retina by pupillary constriction (miosis).
- Visual acuity: ability to look details and contrast. Visual acuity is maximum at the central part of the retina (fovea centralis) where high population of cons is found. 20/20 is normal. The numerator stands for the subject (patient) and the denominator for an ideally normal eye. 20/200 means characters which can be seen by ideally normal eye from 200 feet, can only be seen from 20 feet by a patient. This means the patient is shortsighted (can't see from far). This occurs because of different reasons but most commonly due to a lens shape problem... in shortsighted patient, the image is created at the front of the retina.

RETINA and its PHOTORECEPTOR CELLS (cons and rods)

- There are many differences between cons and rods. IN LIGHT sensitivity, photo pigments and length and number: rods have more but in number of wave length visual acuity and color vision it is con. Cons are more in center but rods are more in periphery.
- The light and photo pigment of the eye is called the rhodopsin. It is composed of a retinal (derivative of vitamin A) and protein opsin. The rhodopsin (the photo and light pigment) helps to absorb light.
- As discussed before, the photoreceptor cells (cons and rods) act as transducers which convert the environmental light energy into electrical energy. This conversion requires a cascade of steps and a signal transduction pathway.

- Steps in the signal transduction pathway (1) Light absorbed by photo pigment (rhodopsin) □ (2) light splits rhodopsin into opsin and retinal □ (3) This induces activation of transducin (a g-protein) □ (4) activated transducin in turn activates a phosphodiesterase which breaks down cGMP □ (5) decreased cGMP □ (6) decrease in cGMP closes Na⁺ ion channel (resulting in Hyperpolarization) □ (7) under hyperpolarized state, voltage gated calcium channels stay closed (calcium can't induce exocytosis or release of inhibitory neurotransmitter) □ (8) Inhibitory neurotransmitter is no longer released from rods □ (8) nerve impulse (AP) is created and message enters the thalamus (particularly the Lateral Geniculate Nucleus – LGN of the thalamus) through optic nerve (CN-II). LGN is an area in the thalamus that integrates visual information (Light). From there information enters the occipital lobe (Visual cortex i.e. area 17 and then to the visual association cortex).

- Breaking down of photo pigment (rhodopsin) helps for light adaptation.
- Re-synthesis of rhodopsin helps for dark adaptation.
- Retinal Cell Layers: The retina is formed by two layers namely:
 - the pigment layer and
 - the receptor cell layer – in turn formed by 2 layers: bipolar & ganglionic

Bipolar cell layer: Consists of bipolar (BP) cells: cells with two processes extending from the soma (cell body). The bipolar cells form synapse both with the cons and rods and the ganglionic cells and hence act as connectors. Even though both cons and rods form synapses with the bipolar cells, the association is different. In case of cons, the association is 1-1 & in case of rods, it's 3-1. 2 rods with 1 Bp

Ganglionic cell layer: Consists of ganglionic cells: cells which produce or generate an impulse or AP. (no AP is created at the level of photoreceptors). The axons of Ganglionic cells, collectively, form optic nerve (CN-II).

- Horizontal and amacrine cells: are involved in lateral inhibition. They modify the responses of bipolar and ganglionic cells via lateral inhibition.
- The Visual neural pathway: photoreceptor cells (cons and rods) □ BP cells □ ganglionic cells (form an AP or action potential) □ LGN □ Visual cortex (area 17) □ Visual association area
- The LGN gets an information not only from ganglionic cells, but also it receives non-retinal inputs for modulation from reticular area and cerebral cortex.
- Tonotopic Representation: Each visual field (right or left) is mapped on a separate (particular) area in the Retina, in separate area in the Lateral Geniculate Nucleus (LGN) of the thalamus and hence in a separate area of the visual cortex as well. First, note that the right temporal part of retina receives left visual field, the left temporal part of retina receives right visual field, the right nasal part of retina receives the right visual field and the left nasal part of the retina receives the left visual field. The LGN receives ipsilateral as well as contralateral

projection from both eyes (because of incomplete decussation of fibers). Fibers from the nasal part of both retinas decussate at the optic chiasma but fibers of the temporal part of the retina go to respective LGN without decussation. This is what incomplete decussation means.

- The Bottom Line is the right LGN & right occipital lobe (right visual cortex) receive visual input from both eyes but from the left visual field only.
- Color Vision: cones are important for color vision because the cones have pigments which correspond to the 3 primary colors of the eye namely – the Red, Green and Blue (RGB). Hence, there are red-sensitive, green-sensitive and blue-sensitive cones. When the subject sees a blue color, the blue-sensitive cones are activated. Respective color blindness results as the respective cones population is reduced.

Hearing and Equilibrium

- outer and middle ear simply conduct a mechanical sound energy to the inner ear which converts the mechanical sound energy to an electrical impulse. Because of this, the outer and middle ear are together known as the conducting apparatus while the inner ear is called the perception apparatus. The outer ear includes the pinna, auditory canal and eardrum (tympanic membrane). The pinna collect sound and direct it to eardrum which in turn conducts the sound to the middle ear. The middle ear mainly bony ossicles (malleus, incus and stapes) which amplify sound to the inner ear. The inner ear contains cochlea (organ of hearing) and otolith organ and semicircular canals (involved in balance).
- Additionally, ceruminous glands found on the wall of external auditory canal produce ear wax known as cerumen.
- In the cochlea, there are receptors along the whole length of the basilar membrane. These receptors aren't all the same to be activated by the same frequency of sound. A high frequency sound activates receptors found on the base of the basilar membrane (this is related to the stiffness of the basilar membrane near its base). As one moves toward the apex of the basilar membrane, the membrane becomes weaker; hence, the receptors near the apex of the basilar membrane are activated by a low frequency sound. The same type of tonotopic representation that is found in the eye exists in the ear. Each segment on the basilar membrane is mapped on a particular area in the temporal lobe.
- The tympanic membrane (eardrum) is a compressible structure b/n outer and inner ear. It has a function of amplifying sound to 10-15 decibel (acts like a lever system along with the ossicles). It may be inflamed and may rupture in a condition called otitis media. This mostly occurs in children and is associated with an URT infection. The tympanic membrane is also affected by external sound intensity. Extremely intense sound may rupture it. The pressure on the tympanic membrane needs to be equalized and this is done by the Eustachian tube. Air conduction is affected by (1) muscles of ossicles (i.e. whether the muscles are contracted or relaxed). When the intensity of sound is high, the muscles of the ossicles contract. This contraction of muscles decreases the vibration of the bones and hence decreases the high vibration energy that passes into the inner ear. This is a natural mechanism to prevent an inner

ear damage and is called attenuation reflex (stapedius reflex) – an example of an auditory reflex. Air conduction is also affected by the ability of (2) bone conduction (ability of the ossicles to conduct sound). The ossicles of the middle ear may sometimes lose their ability to conduct sound. This results in a conduction deafness and can be corrected by a hearing amplifier (aid) placed on the temporal lobe to bypass the amplification by the ossicles. Know the factors that affect conduction well such as the ossicles (the bones themselves), their muscles, ligaments, and also presence of abnormal conditions such as extra-bony structure in the ear.

- The cochlea has fluid filled membranes namely: scala tympani, scala vestibuli and scala media. The scala tympani and vestibuli contain perilymph while the scala media contains endolymph (contains more K^+ and less Na^+ like the ICF). In all other body parts, except the scala media, opening of K^+ channels, drives out K^+ because of the concentration gradient (higher concentration of K^+ is found inside cells). The scala media are the only exceptional cells where an opening of K^+ facilitates influx of more K^+ against its concentration gradient. Know this important exception.
- Other than the 3 fluid filled membranes, the cochlea also has a basilar membrane. Remember the basilar membrane is stiff at base for high frequency sound and flexible at the apex for a low frequency sound. This is called place principle, Because the sound frequency should be high to activate the receptors on the stiff side of the basilar membrane, a low frequency sound passes to the apex of the basilar membrane without activating the receptors on the stiffer (basal side) of the basilar membrane. (look at slide 45 or 51 for an image). Sound intensity means the loudness of a given sound. Sound intensity is determined by the amount of NT released, number of cochlear nerve stimulated, number of receptors activated, types of receptors activated and frequency of AP. For example, the higher the sound intensity (loudness), the higher the frequency of AP that is transmitted to the brain.
- Organ of Corti: a structure sitting on the basilar membrane. Organ of Corti contains receptors – mechanoreceptors which detect sound energy (special cell receptors). These receptor cells are called the Hair cells. The hair cells have stereocilia which attaches them to a fixed tectorial membrane. When a vibration of fluid is generated, the vibrating basilar membrane pushes the hair cells upward against resistant/fixed tectorial membrane. This creates a pressure on the hair cells which opens the K^+ ion channels and K^+ to leak in (an important exception). This triggers depolarization resulting in Ca^{2+} influx (due to opening of voltage gated calcium channels). Ca^{2+} facilitates the exocytosis or release of the neurotransmitter glutamate from the vesicle of the hair cells.
- Auditory Pathway: Cochlear nerve (formed by the axons of afferent neurons in organ of Corti which generate the AP) transmits auditory impulses from the organ of Corti to spiral ganglion. Glutamate released by the cochlear nerve ending activates the spiral ganglion. From spiral ganglion, vestibulocochlear nerve (CN-VIII) carries the auditory information to brain. On its way to brain, the information enters to two auditory nuclei in the medulla – the cochlear nucleus and the superior olivary nucleus. Superior olivary nucleus helps to identify the exact origin (& direction) of a certain sound while inferior olivary nucleus is involved in auditory reflex. Auditory reflex is any response that is manifested following an auditory nuclei activation.

The sensory fibers of the auditory pathway decussate at the medulla as they ascend to thalamus via the inferior colliculus. The auditory information is, then, integrated at the level of the thalamus in the Medial Geniculate Nuclei (MGN) of the thalamus. [Auditory=Music goes to MGN. Visual=Light goes to LGN.] After the auditory information is integrated in the thalamus, it enters the primary auditory cortex in the thalamus.

- As mentioned before, the superior olivary nucleus is important for detecting the direction of sound origin. This detection is based up on (1) the intensity of the sound (the higher the intensity, the easier the detection of origin). (2) the arriving time (the sooner the sound arrives, the easier the detection from where it came from)
- The inner ear not only contains the cochlea but also the vestibular apparatus; and the vestibular apparatus is very important to keep our equilibrium; it is not only cerebellum that is involved in maintaining equilibrium and posture;
- Equilibrium means keeping our balance while we're moving (kinesthetic balance);
- There should be a way through which cerebellum is informed about the body movement and position;
- The vestibular apparatus in the inner ear are important either for linear acceleration (those which we call otholith organs) or for angular acceleration (the semicircular canal);
- In addition, there should be reflexes, which are important to maintain posture – called postural
- reflexes;

Types of equilibrium

- Static equilibrium – maintainance of the body position relative to gravity;
- Dynamic equilibrium – maintainance of body posture in response to some kind of movement; When there is some kind of movement or when a person is about to fall down, cerebellum detects it; the intention that is planned by the motor area of the cerebellar cortex normally gets into the cerebellum; info about our “current” or present movement is also sent to the cerebellum; thus, the cerebellum contains the intension that is formulated by the motor area and the “current” movement,if there is mismatch, there should be correction; meaning, if there is a mismatch between what the brain think (intended) and our movement, there should be correction; this is called the cerebellocomparator function of the cerebellum;
- So in order for the cerebellum to do the above, it should be informed about the body position; and this is done through the proprioceptive system (by the receptors in joints); even when we are moving our body in a certain direction we can have the fluid in the inner ear move to opposite direction => when this happens, the hair cells will bend => then the message gets into the vestibular nuclei in brainstem => finally gets into cerebellum;

- Vestibular nuclei are groups of neurons found in the brainstem that receive info from vestibular apparatus; they send a message into the cerebellum; when information reaches to the nodular part of the cerebellum, dynamic equilibrium is developed; we can also have this area of cerebellum for static equilibrium (may be showing on power point); once info has reached to cerebellum, different areas are involved in maintenance of equilibrium, not just one area;
- Semicircular canals are involved in dynamic equilibrium; lesion results in abnormalities in dynamic equilibrium;
- Semicircular canals contain ducts, and these ducts contain fluid;
- Utricle and saccule are important for linear acceleration; and the semicircular canal is important for angular acceleration; these 3 are what we collectively call vestibular apparatus;
- the ear is not only important to hear sound, but also to keep balance;

organs of static equilibrium

- the saccule and utricle are collectively called otolith organ; they are important for linear acceleration; the arrangement of saccule and utricle is not the same; their arrangement is important to detect different types of movements; the utricles are arranged vertically and they detect horizontal movement of the subject;
- the maculae structure, found in the otolith organ, contains receptor cells that detect movement; and it detects linear or static movement/acceleration; the cells, in this structure, are arranged in different ways so that they have a potential to detect different types of movements; also, cells in the maculae sense the movement of the head with respect to gravity; when the head is tilted, the heavy gel-like structure found in the maculae (which has high density) bends the receptors towards the long hair found on the top of the receptors => then potassium channels open and potassium gets into the cell=> depolarization occurs=> the depolarization facilitates the calcium channels to be open => calcium gets into the cell => calcium activates synaptic proteins (proteins found at synapses that are important for docking – attachment of active vesicles with axon terminal membrane) => exocytosis => NTs are released => impulse is generated and gets to

vestibular nuclei in the brainstem => there will be integration here => finally the info gets into cerebellum;

- when the head is upright, the receptors (hairs) are also upright/not bent; when the head is tilted to the side, these receptors bend, and this bending is important for the opening of potassium ion channels;
- potassium is very important for hearing and balance; because it is found in the endolymph;
- All the potassium channels are mechanically attached to each other (by a rail-like protein); thus, when one potassium channel opens, all the other channels will also open;
- The special cell receptors don't have an action potential that can be created; so the graded membrane potential/receptor potential is developed; and the receptor potential facilitates the release of NT; then the NT will be attached to receptor found on postsynaptic cell => action potential will be created if depolarization reaches to critical value or threshold membrane potential => then the impulse gets into the brain stem;

Dynamic movement

- The ampullae structure contains cell receptors detecting dynamic movement; this structure is found within the semicircular canal; there are sensors in this structure – they are mechanoreceptors, because they respond for mechanical activities; is a structure found in the semicircular canal that detect angular movement; Even the eye is very important for the vestibular system; the visual system is involved in the maintenance of our posture; once info is integrated we have the oculomotor nerve that moves the eye; for eg. If we move our head in a certain direction, our eye should move to that direction so that our vision can be fixed on the target we intended (fixation), for this there should be integration;

THE CHEMICAL SENSES

Taste & Smell allow us to separate pleasant & nutritious foods from undesirable (lethal) foods.

Impulses for smell & taste propagate to limbic system (higher cortical areas),

—certain odors & tastes can evoke strong emotional responses or a flood of memories.

In both smell and taste chemoreceptors respond to chemicals in aqueous solution

Taste—to substances dissolved in saliva

Smell—to substances dissolved in fluids of the nasal membranes

Olfactory epithelium occupies superior part of nasal cavity & consists of 3 kinds of cells:
Olfactory receptors, Supporting cells & Basal cells.

Olfactory Pathway...

- On each side of nose 40 bundles of axons of olfactory receptors form right & left olfactory nerves & terminate in olfactory bulbs (Organ which houses all the nerves which receive inputs from the olfactory receptors)

- axon terminals of olfactory receptors form synapses with olfactory bulb neurons. Axons of olfactory bulb neurons extend posteriorly & form the olfactory tract.

- Some axons of olfactory tract project to 1^{ry} (primary) olfactory area of cerebral cortex (temporal lobe), where conscious awareness of smell begins.

Olfactory sensations are the only sensations that reach the cerebral cortex without first synapsing in the thalamus.

- Other axons of olfactory tract project to limbic system & hypothalamus (emotional & memory evoked responses to odors). sexual excitement on smelling certain perfume, nausea on smelling food that once made you violently ill, or odor evoked memory of a childhood experience.

- From 1^{ry} olfactory area, pathways also extend to the frontal lobe (orbitofrontal area/ area 11). odor identification & discrimination area.

Olfactory mucosa is directly linked to olfactory bulb which sends the information to the frontal lobe for identification

- Collaterals are sent to three other structures Amygdala—where emotions are processed
Hippocampus—where memories are processed Hypothalamus—processing involved in eating, drinking & sexual behavior.

- Smell is strongly associated with emotions, memory, eating, drinking & sexual behavior.

Very Old, Less Old & Newer Olfactory Pathways into CNS

- Medial olfactory area represents very old olfactory system. And Lateral olfactory area is input to less old & newer system

Very old olfactory system: subserve basic olfactory reflexes (lips licking, salivation & other feeding responses caused by smell of food or by primitive emotional drives associated with smell).

Less old system: provides automatic but partially learned control of food intake & aversion to toxic & unhealthy foods.

Newer system: comparable to most of other cortical sensory systems & is used for conscious perception & analysis of olfaction

In addition to basic chemical mechanism by which olfactory cells are stimulated, several physical factors affect degree of stimulation. only volatile substances that can be sniffed into nostrils can be smelled. stimulating substance must be at least slightly water soluble so that it can pass through mucus to reach olfactory cilia. it is helpful for the substance to be at least slightly lipid soluble, presumably because lipid constituents of cilium itself are a weak barrier to non-lipid-soluble odorants.

Olfaction has low threshold, only few molecules of certain substances need be present in air to be perceived as an odor. Eg: methyl mercaptol smells like rotten cabbage is detected 1/25 billionth of a mg/ml that's y its put in gas used for cooking.

Adaptation(decreasing sensitivity) to odors occurs rapidly. olfactory receptors adapt by about 50% in 1st second or so after stimulation but adapt very slowly thereafter. complete insensitivity to certain strong odors occurs about a minute after exposure. There is adaptation process in CNS. Large numbers of centrifugal nerve fibers pass from olfactory regions of brain backward along the olfactory tract & terminate on special inhibitory cells in olfactory bulb (granule cells). after onset of olfactory stimulus, CNS quickly develops strong feedback inhibition to suppress relay of smell signals through olfactory bulb.

Amoores 7 basic smells are found and all other smells would be a combination of these

Humans are microsmatic i.e hav less keen sense of smell than animals (microsmatic) b/c we have much less number of receptors.

Smell has more affective in determining pleasantness of food than taste. Smell plays major role with emotion and sexual drive

Pheromones: Chemical secreted by animals having; Powerful effects on behavior, specifically sexual behavior, territorial behavior and identification of kin. In Mammals, Most mammals only become sexually aroused in the presence of pheromones. Pheremons also have major effect on humans.

Gustation.

5 primary tastes can be distinguished; sour, sweet, bitter, salty and umami(meaty/savory) arise from taste receptors that are stimulated by monosodium glutamate (MSG).

All other flavors, are combinations of 5 primary tastes, plus accompanying olfactory & tactile(touch by tongue) sensations.

Taste is mainly a function of taste buds in mouth, but it is common experience that one's sense of smell also contributes strongly to taste perception. Taste is 80% smell. Taste allows selection of food according to desire or tissue metabolic needs. when you have a cold or are suffering from allergies & cannot taste your food, it is actually olfaction that is blocked, not taste.

Taste receptors are located in taste buds some on tongue some on palate pharynx and epiglottis

Taste bud has: Supporting cells, Gustatory receptor cells & Basal cells.

TASTANTS= chemicals that stimulate gustatory receptors. once tastants dissolved in saliva it can make contact with plasma membrane of gustatory hairs (sites of taste transduction). Receptor potential stimulates exocytosis of synaptic vesicles from the gustatory receptor cell. liberated NT molecules trigger nerve impulses in 1st order sensory neurons that synapse with gustatory receptor cells.

Dissolution in saliva-> attachment to receptors-> generator potential-> action potential

Different ions, different receptors for different tastes.

Na in salty food enter gustatory receptor cells via Na channels in the plasma membrane. accumulation of Na inside causes depolarization that leads to release of NT.

- H^+ in sour tastants may flow into gustatory receptor cells via H channels. they also influence opening & closing of other types of ion channels & result in depolarization & liberation of NT.

Tastants stimulating sweet, bitter & umami tastes, do not themselves enter gustatory receptor cells. bind to receptors on plasma membrane that are linked to G proteins. G proteins then activate several different chemicals known as 2nd messengers inside gustatory receptor cell. different 2nd messengers cause depolarization in different ways, but result is the same - release of NT.

If all tastants cause release of neurotransmitter, why do foods taste different? b/c patterns of nerve impulses in groups of 1st order taste neurons that synapse with gustatory receptor cells.

Different tastes arise from activation of different groups of taste neurons. although each individual gustatory receptor cell responds to > 1 of 5 primary tastes, it may respond more strongly to some tastants than to others.

Gustatory pathway: 3 cranial nerves contain axons of the 1st order gustatory neurons that innervate the taste buds. Facial nerve from anterior 2/3 of tongue. Glossopharyngeal nerve „„posterior 1/3 of tongue and Vagus nerve „„throat & epiglottis. From taste buds, nerve impulses propagate along these cranial nerves to gustatory nucleus in medulla oblongata. From medulla; some axons carrying taste signals project to limbic system & hypothalamus and others project to thalamus. Taste signals that project from thalamus to 1st gustatory area in parietal lobe of cerebral cortex (area 43) give rise to conscious perception of taste.

Taste Reflexes Integrated in Brain Stem: From the tractus solitarius, many taste signals are transmitted within the brain stem itself directly into the superior and inferior salivatory nuclei, and

- these areas transmit signals to the submandibular, sublingual & parotid glands to help control the secretion of saliva during the ingestion & digestion of food.

Threshold for bitter is lowest b/c most poisons are bitter it has protective function b/c intensely bitter taste leads to rejection of food. Threshold for sour (in spoiled food) taste is somewhat higher than bitter. Thresholds for salty substances & sweet substances are similar & are higher

than those for bitter or sour substances. Complete adaptation to specific taste can occur in 1-5 minutes of continuous stimulation. Taste adaptation is at level of receptor and also CNS.

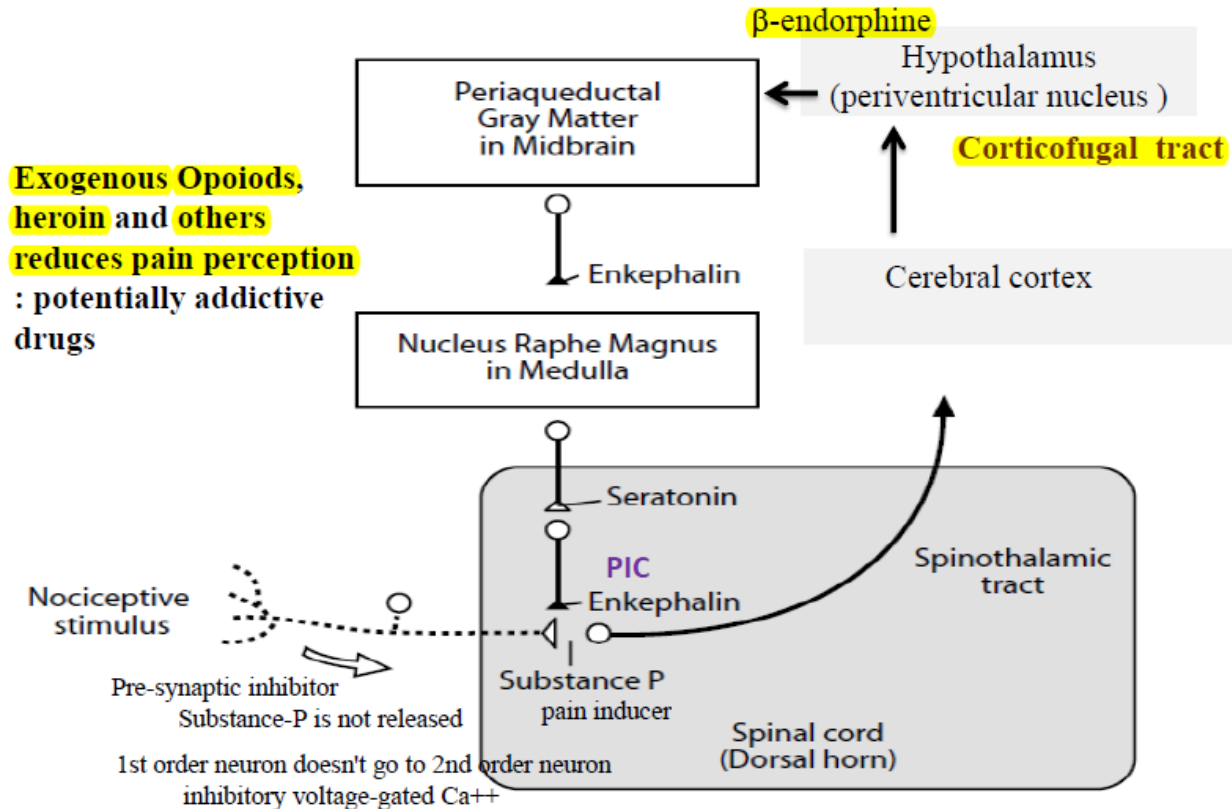
Animals use taste preference (choosing certain types of food than others) to control type of diet they eat. Taste preferences often change in accord with the body's need for certain specific substances: adrenalectomized people (salt depleted) select water with high NaCl than pure water. Insulin injected choose sweet food. Parathyroidectomy patient like drinking water with high CaCl.

The disorders of smell are classified as "-osmias" and those of taste as "-geusias."

PAIN MODULATION

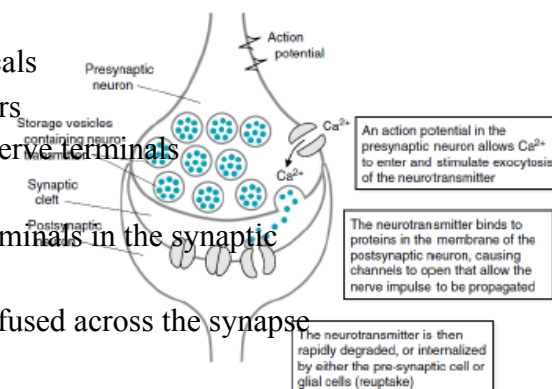
- Pain we perceive is not always the same; it is dynamic
 - Analgesia → Decrease Pain
 - Hyperalgesia → Increase pain
- Hyperalgesia (Elevated pain perception)
 - Primary: When the pain receptors are sensitized by chemicals produced by injury
 - Bradykinins, histamine, serotonin, leukotriene, prostaglandins, substance-p and cytokines → Induce pain
 - Reduce receptor threshold
 - Secondary: When the pain area/final neurons or neurons along the pain pathway are sensitized
 - Hyperpathia: Excessive sensitivity and raised threshold to painful stimuli
 - Dorsal horn neuron **infection** (e.g. Herpes virus: Increase pain sensitivity)
 - Strong/intense pain, rash skin, red → on dermatome
 - Action potentials in sensory fibers and all its branches release substance-p from injured area
 - Reddening/flare is due to collateral neurons dilating blood vessels
 - Secondary hyperalgesia is due to neurotransmitters in the spinal cord and brain
 - Allodynia (very severe) → Causes non-pain stimulus (e.g. wearing tight cloth or cold showers) to be painful
 - Tic Douloureux (a.k.a Trigeminal neuralgia)
 - Half your face is in pain (intense/chronic) associated with trigeminal and glossopharyngeal nerve problems
 - Electric shock or lancinating type of pain
 - Causalgia
 - Severe burning pain in limbs via **injury to peripheral nerve**
 - Usually felt in hands and feet; Chronic
 - Diabetes → Glycated end products effect the end of nerve endings (especially finger)
 - Hyperpathia (result of injury) / Increased sympathetic effectors

- Hyperalgesia; Allodynia; Injured nerve develops branching fibers and stimulate the sensory fibers from the injured area
- Phantom pain (Feel pain in area of amputated limb that no longer exists)
 - Inflamed sensory nerve ending at the amputated end
 - Axonal sprouting (Branching of collateral nerve) and maladaptive plasticity
- Analgesia
 - Occurs at different levels
 - At receptor levels → e.g. Aspirin inhibits action of Prostaglandin
 - At spinal cord level → pain-gated area → block gate
 - Opiate drugs act on CC, Thalamus, Brain-stem, Spinal cord
 - Peripheral analgesic system
 - Dorsal horn spinal gate → Pain inhibiting complex (PIC) is located
 - Non-painful input closes the gate to painful input at the spinal cord level
 - Peripheral tactile receptors and pain fiber → Interneuron
 - Touching area around pain/injury (activates peripheral tactile receptors) → Decreases intensity of pain via forward inhibitory
 - Descending central analgesic system



NEUROTRANSMITTERS and NEUROTOXINS

- Facilitate cell-cell communication at the synapse via chemicals
- Synthesis from amino acid and common metabolic precursors
 - Usually occurs in the cytoplasm of the presynaptic nerve terminals
- They are produced in the cell body
 - Transported to the ends of the axon (presynaptic) terminals in the synaptic vesicles (where they are stored)
 - In response to the action potential signal they are diffused across the synapse
 - They act at a receptor on the post-synaptic membrane
 - Most neurons can take in 3-5 different types of neurotransmitters
 - Their action is terminated through reuptake into the pre-synaptic terminals, diffusion away from the synapse, or enzymatic inactivation
- Neuronal SNARE (Soluble N-ethylmaleimide): Mediate membrane fusion in synaptic vesicle fusion/fuse synaptic vesicles w/ plasma membrane. 3 neuronal SNARE proteins
 - Syntaxin, SNAP25, Synaptobrevin
- Families of neurotransmitters: Cholinergic agents, catecholamines, amino acids and derivatives, peptide neurotransmitters, gaseous neurotransmitters



Acetylcholine (Cholinergic Synapse)

- Arriving action potential depolarizes membrane → open Ca^{++} channels → vesicles fuse with plasma membrane → Ach spills into cleft → migrate to adjacent cells and binds to receptors
- Responsible for stimulation of muscles; Derived from Choline; Functions in both CNS and PNS
- Toxin effects
 - **Botulism** toxin: Inhibits Ach release
 - Predominantly affect the peripheral nervous system at a neuromuscular junction. Blocks the release of acetylcholine at synaptic cleft, which is required for a nerve to stimulate the muscle-blocking muscle stimulation resulting **flaccid paralysis**
 - **Lactrotoxin** (Black widow's): Protein overstimulation
 - Acts presynaptically to release Ach from sensory and motor neurons (as well as endocrine cells (insulin))
- Receptor
 - Nicotinic (voltage-gated ion channels/ionotropic)
 - Muscarinic (G-proteins/metabotropic)
 - Antagonists of muscarinic Ach receptor → Atropine
- Degraded via **Acetylcholinesterase** (hydrolysis) in the cleft

- Covalent organophosphorus inhibitors of acetylcholinesterase
 - DIFP(diisopropylphosphofluoridate); Nerve gases (tabun and sarin); Insecticides (parathion and malathion)
- Non-covalent inhibitors of acetylcholinesterase
 - Physostigmine and Neostigmine → used to treat **Myasthenia Gravis**
- Reuptake via transport proteins and V-type H⁺-ATPase
- Synthesis: Choline + Acetyl co-enzyme A by choline acetyltransferase in the cytoplasm
- Loss of Ach (90%) in the brain linked to Alzheimer's

Glutamate (Excitatory)

- Most common NT in CNS; Important for memory and learning; Toxic to neuron in excess
- It is either returned to neuron or carried into glial cells → converted to Gln and taken back to the neuron
- Receptors
 - Ionotropic
 - AMPA receptor
 - Kainate receptor
 - NMDA receptor (Best understood)
 - Ca²⁺ channels open → Mg²⁺ blockade is removed (via depolarization from non-NMDA receptors via Na⁺ channel opening) → Ca²⁺ enters when Mg²⁺ is removed
 - Post-synaptic effect → learning
 - Ca²⁺ dependent K⁺ channels open → repolarization → reinstates Mg²⁺ blockade
 - PCP (phencyclidine/angel dust): Antagonist to NMDA receptor
 - Once used as an anesthetic agent. Psychotic/hallucinogenic
 - Metabotropic
 - mGluRs → there are 8 (mGluR₁ – mGluR₈)
- Synthesized
 - Glycolysis → Krebs cycle → from α-ketoglutarate via glutamate transaminase or dehydrogenase; Glutamine (deamination); Aspartate
- Removal
 - Excitatory amino acid carrier-1 (EAAC1); glutamate transporter-1 (GLT-1); glutamate-aspartate transporter (GLAST)
- Excitatory glutamate is broken down to inhibitory GABA
- ALS (a.k.a Lou Gehrig's Disease) results from excessive glutamate production
 - Amyotrophic lateral sclerosis results in degeneration of motor neurons in the anterior horn of spinal cord, brainstem and cerebral cortex

- Like **aspartate** it is a non-essential amino acid; they don't cross the BBB so they must be synthesized in neurons and glial cells; both excitatory NTs

GABA (gamma-Amino Butyric Acid) and Glycine

- **Inhibitory NTs:** Diminish action of activating NTs
- Activate Cl⁻ channels; Make post-synaptic membranes permeable to chloride ions
 - Influx of Cl⁻ causes hyperpolarization of the post-synaptic membrane → make membrane more negative
- GABA operates mainly in the brain; Glycine acts in **spinal cord** (also brainstem and retina)
- GABA is derived from glutamate
 - Glutamic acid via Glutamic acid decarboxylase (GAD) → GABA
- GABA receptors
 - GABA_A (ionotropic) and GABA_B (metabotropic)
- Glycine receptor is site of action of strychnine/strychnine has a high affinity for glycine receptor (Strychnine blocks inhibitory neurons)
- Glycine is synthesized from serine via serine hydroxymethyltransferase through 3-phosphoglycerate
 - Removed via high-affinity transporter

Diffusible Gases

- Nitric Oxide (NO)
 - Arginine via Nitric Oxide Synthase (NOS) → NO + Citrulline
 - Glutamate is released → acts on NMDA receptors located on the post-synaptic neurons → Ca²⁺ enters the post-synaptic neuron and binds with calmodulin activating NOS
 - NO stimulates guanylate cyclase forming cGMP → physiological response
 - Not stored in vesicles; not released by Ca²⁺ dependent exocytosis (it diffuses); inactivation is passive (decays); it does not interact with receptors on target cells (sphere of action depends on the extent it diffuses and its actions is not confined to presynaptic-postsynaptic direction)
 - Inhibits hemoglobin and other heme proteins which bind it tightly (Half-life 2-4 seconds)
 - NOS (All 3 isoforms require BH₂ as a cofactor and NADPH as a coenzyme)
 - Isoform I (nNOS or c NOS): Neurons and epithelial cells. Activated by influx of extracellular Ca²⁺.
 - Isoform II (iNOS): Macrophages and smooth muscle cells. Induced by cytokines

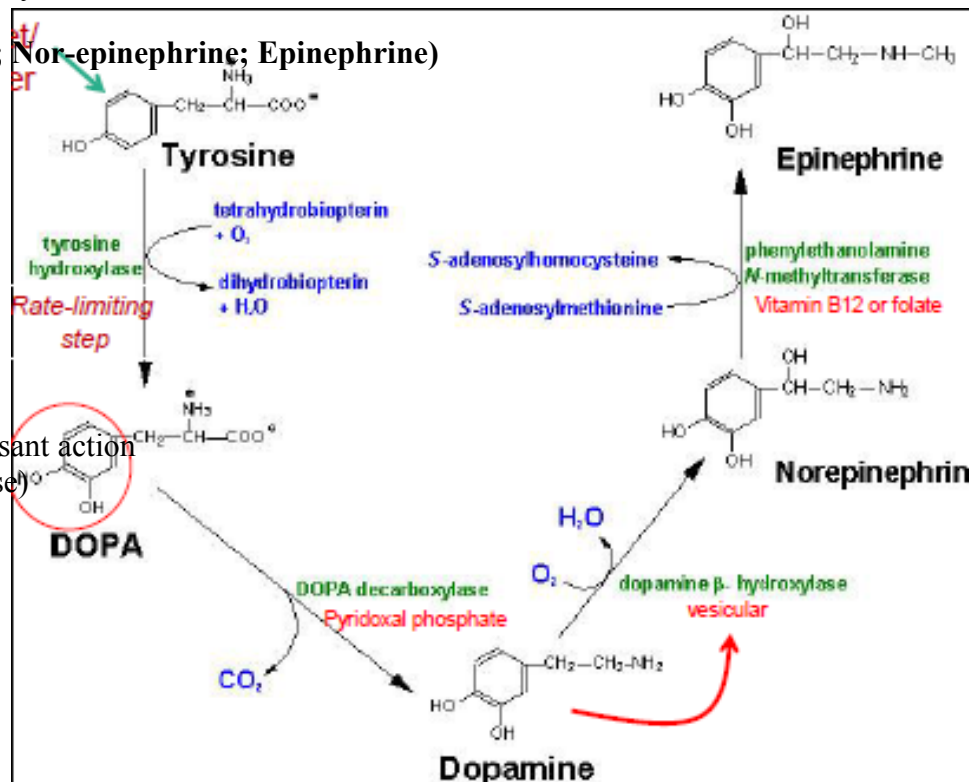
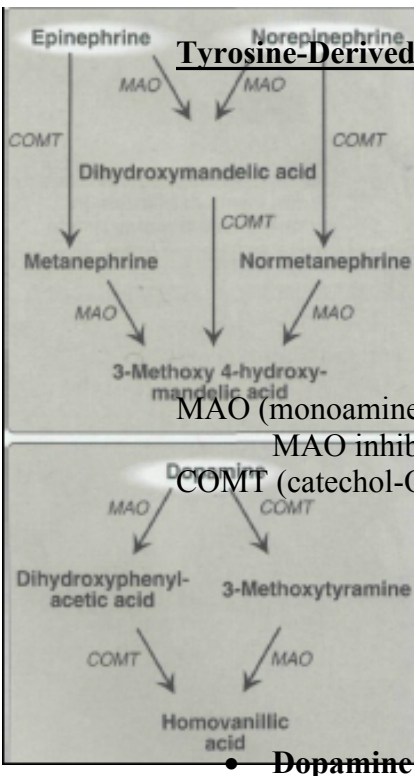
- Isoform III (eNOS): Endothelial cells lining blood vessels. Activated by influx of extracellular Ca^{2+} .
- Carbon Monoxide (CO)
 - Heme via Heme Oxygenase $\rightarrow$ CO + biliverdine
- Regulate blood flow (cerebral and peripheral (e.g. penis))
- Retrograde messenger: Chemical substance that is released from post-synaptic neuron and acts on the pre-synaptic neurons.

Tryptophan-Derived NTs (Serotonin and Melatonin)

- **Selective serotonin re-uptake inhibitors** (SSRIs/Antidepressants/Prozac) $\rightarrow$ inhibit reuptake process resulting in prolonged serotonin presences in the synaptic cleft
- Serotonin
 - Emotion and mood with low concentration of serotonin leading to depression, anger, obsessive-compulsive disorder, and suicide
 - Has a complex modulatory control over glutamate and GABA mediated transmission
- Serotonin is synthesized in pineal gland
 - It is a precursor for melatonin
 - Melatonin: Neuro-hormone which regulates sleep pattern, seasonal and circadian(daily) rhythms; dark-light cycle
 - Tryptophan via TPH $\rightarrow$ 5-HT via AADC $\rightarrow$ Serotonin via SHA (acetyl-CoA) $\rightarrow$ N-acetylserotonin via HMT $\rightarrow$ Melatonin

Tyrosine-Derived NTs (Dopamine; Nor-epinephrine; Epinephrine)

MAO (monoamine oxidase)
MAO inhibitors: Anti-depressant action
COMT (catechol-O-methyltransferase)



- Both excitatory (D1) and Inhibitory (D2) depending on the receptor sites
- Strongly associated with reward mechanism in the brain (if it feels good dopamine neurons are probably involved) → addiction
- Schizophrenia → Excessive dopamine in the frontal lobe
- Parkinson's Disease → Depletion of dopamine in the motor areas of the brain
- Epinephrine and NE bind to α and β –adrenergic receptors (GPCRs)
- **Disulfiram**
 - Inhibition of Ethanol Metabolism
 - Inhibits aldehyde dehydrogenase (ALDH) which is needed in order to transform acetaldehyde into **acetate** (Ethanol is 1st converted to acetaldehyde via alcohol dehydrogenase)
 - Basically it prevent people from drinking because the inability to convert ethanol to acetate makes you vomit.
 - Inhibition of Norepinephrine
 - Disulfiram inhibit dopamine beta-hydroxylase (Look at picture above)
 - Used in treatment of cocaine dependence
- **Amphetamines** inhibit tissue reuptake catecholamines
 - Lower brain levels of NE via releasing of NE from vesicles
 - Dopamine → Increase concentration of Dopamine
 - 1. Reversal of DA uptake carrier 2. Interference with uptake into DA vesicle 3. Inhibition (at high concentration) of MOA

Histamine

- Does not penetrate BBB → hence must be synthesized
- Derived from histidine and produced by mast cells, basophils, enterochromaffine like cells (ECL) of the stomach, and hypothalamus

Peptide NTs

- Low concentration; purification is hard; complex roles
- Endorphins and enkephalins are natural opioids/pain killers
- Endothlins affect smooth muscle contraction, vasoconstriction, mitogenesis (induction of mitosis), tissue change
- Neurotrophins induce survival, development, and function of neurons (growth and maintenance of neurons)
 - Receptors → p75 and Trk (Tyrosine kinase receptors)
 - Brain derived neurotropic factor (BDNF) → learning and memory
 - Plasma BDNF → biomarker of impaired memory and cognitive aging for women
 - Hippocampal samples from donors with Alzheimer's show decrease of BDNF

- BDNF increases phosphorylation of AMPK and acetyl CoA carboxylase (ACC) and enhance the rate of fatty acid oxidation
 - BDNF → pAMPK → pACC → Fat oxidation

PROGRESSIVE NEURODEGENERATIVE DISEASES

Parkinson's Disease (1% of people over 65)

- Caused by degeneration of nigrostriatal system → the dopamine secreting neurons of the substantia nigra (dopaminergic neurons die)
 - Loss of dopamine-containing cells in nigrostriatal pathway → Motor disorder
- Lewy Body: abnormal circular structure with a dense core consisting of **α -synuclein** protein (pre-synaptic protein); found in dopaminergic nigrostriatal neuron of people with parkinson's
- Symptoms: Muscular rigidity, slowness of movement (bradykinesia)/difficulty initiating movement (Akinesia); resting tremors; postural instability; difficulties with handwriting or making facial expression; shuffling gait; memory loss
 - Advance stage → psychiatric complication (depression; hallucination; paranoia)
- Clinical Dx usually made after 80% of striatal dopamine is lost; symptoms after 60%
- More common in women than men
- Most cases unexplained(primary/idiopathic); Secondary → drugs, stroke, or trauma ("Frozen addict" → impure heroin kills dopaminergic cells); influenza; Genetics (PARK gene 1-10 and 16)
 - Smoking protects against PD*
- Gene Mutation of Chromosome 4
 - Gene that code for α -synuclein (SNCA/ located in presynaptic terminal of DA cells).
 - Dominant. Abnormal SNCA becomes misfolded, forms aggregation (make up lewy bodies)
- Mutation of Chromosome 6 forms abnormal Parkin protein. Recessive.
 - Normal Parkin traffic defective/misfolded proteins to proteasomes for destruction
 - Abnormality → high levels of defective proteins accumulate in dopaminergic neurons → kills cells
- Toxic chemicals (e.g. insecticides) inhibit mitochondrial function with leads to aggregation of misfolded SNCA in DA neurons → kills cells
- Why the nigrostriatal (NGS) pathway?
 - Ca²⁺ channels regulate spontaneous activity of DA cells in NGS pathway
 - Na⁺ channels regulate spontaneous activity of DA cells in other pathway
 - In animal models it was shown
 - α -synuclein (SNCA) is increased via Ca²⁺ which in turn increases the death of DA cells

- Since DA cells in other pathways do not contain increased Ca^{2+} → neurons are spared
- Tx:
 - Levodopa (L-DOPA) usually combined with carbidopa (peripheral decarboxylase inhibitor)
 - Causes remaining DA neurons to secrete more DA → alleviates symptoms
 - Very effective for the first 2-5 years of treatment after which the on-off effect develops due to sensitization or continued loss of dopamine cells
 - Direct-acting dopamine agonists
 - Experimental brain surgery
 - Pallidotomies: surgical destruction of internal division of globus pallidus
 - Helpful to know that decrease in activity of DA input to caudate nucleus and putamen cause an increase in activity of globus pallidus
 - Neural tissue implants (e.g. fetal dopamine cells)
 - Electric stimulation or lesion of subthalamic nucleus
 - Subthalamic nucleus has excitatory effect on globus pallidus and damaging it decreases such actions
 - Gene Therapy
 - Genetically modified virus into subthalamic nucleus
 - Delivers a gene for GAD (enzyme that makes GABA) → production of GAD turns some glutamate neurons into inhibitory GABA neurons → activity of globus pallidus is decreased → activity of supplementary motor area increased and symptoms improve

Alzheimer's Disease

- Begins with a “pure” impairment of cognitive function/Decline in Ach levels
- Progression (slowly)
 - Mild forgetfulness (names; recent events; simple math; repeat themselves) → Forget how to do simple task (brushing teeth/hair); no longer think clearly; problems speaking, reading, writing, or understanding → Late stage: anxious or aggressive; dementia; wander from home; eventually patient need total care
 - Eventually coma and death (usually from infection)
- Most common AD is late-onset or sporadic AD (occurs late in life and has no obvious inheritance pattern)
 - However, several risk factor genes may interact with each other causing late onset AD
 - Most common risk factor gene ApoE4 (makes apolipoprotein E)
- Familial AD is rare (start at age 30-60); Abnormality on chromosome 21
 - All known genetic risk factors increase accumulation of $\text{A}\beta$ peptides

- Anatomical Hallmark → **Amyloid plaque** and **Neurofibrillary tangles** present in CNS (initially in the hippocampus cells degenerate and later in the cerebral cortex) → CNS atrophy
 - **Amyloid plaque** contain large amounts of 42 amino acid peptide termed **β -amyloid ($A\beta$ -42)**/formed by defective β -amyloid protein ($A\beta$)
 - Insoluble deposits intermingled with portion of neurons and with microglia and astrocytes
 - Extracellular deposit containing a dense core of β -amyloid protein surrounded by degenerating axon and dendrites and activated microglia and reactivate astrocytes
 - Hypothesis: β -amyloid itself initial causes pathophysiology that leads to dementia.
 - Gene encodes the production of β -amyloid precursor protein (APP)
 - APP is then cut in 2 places by secretase to produce β -amyloid protein → Results in $A\beta$ -40 and $A\beta$ -42
 - In normal brain → 95% is $A\beta$ -40(short)
 - In AD → $A\beta$ -42 (long) is as high as 40% (folds improperly and forms aggregates)
 - Aggregated forms of amyloid
 - Interact with microglia causing inflammatory response that triggers the release of toxic cytokines
 - Triggers release of glutamate by glia cells → causes excitotoxicity
 - Cause synaptic dysfunction and suppress the formation of LTP
 - **Neurofibrillary tangles** is rich in cytoskeletal protein (especially microtubule-associated protein named **tau**) → twisted microfilaments with abnormal Tau
 - In the tangles → heavily phosphorylated proteins which may cause aggregation and precipitation of the cytoskeleton
 - Unclear whether β -amyloid and neurofibrillary tangles cause AD or are products of the disease
- Tx
 - Acetylcholinesterase inhibitors (Aricept; Exelon; Reminyl)
 - Because cholinergic basal forebrain neurons are among the first to die
 - NMDA inhibitors (Memantine)
 - Growth factors

Huntington's Disease

- Caused by an autosomal dominant mutation of a gene that codes **Huntingtin** protein (htt) (50% chance of inheriting the disease)
 - Chromosome 4
 - Repeated sequence of bases that code for the amino acid glutamine

- Abnormal htt becomes misfolded and forms aggregates in nucleus → cell death (apoptosis)
 - Abnormal htt has long glutamine tract → may lead to abnormal endocytosis and secretions of neurotransmitters
 - Striatal death by apoptosis (caspase-3)
 - Abnormal htt interferes with BDNF in both gene expression and transport from cerebral cortex to basal ganglia
- Degeneration of caudate nucleus and putamen
 - Neurodegeneration in **Putamen**
 - 1st loss of inhibitory neurons (GABAergic)
 - Removes inhibitory controls of motor areas in cortex (hyperkinetic)
 - As the disease progress → occurs in many other regions
- Uncontrollable movements, jerky limb movement
 - Physical symptoms can begin at any age but usually begin btw 35-44
 - 6% start before 21 years with akinetic-rigid syndrome (progress faster)
- Death (10-15 years); Progressive cognitive and emotional changes
- Tx → None
 - Happ1 –Antibody that acts intracellularly (intrabody)
 - Targets a portion of the htt protein
 - Mouse Models → insertion of Happ1 gene into brain suppressed production of mutant htt and improved symptoms

Multiple Sclerosis

- Autoimmune demyelinating disease
 - Patchy loss of myelin in areas of the brain and spinal cord
 - Affects only oligodendrocytes not Schwann cells
 - Significant neural recovery often occurs during periods of **spontaneous remission** (sometimes > 10 years)
 - Conduction of action potential slowed and Na⁺ channels spread outward from the nodes lowering their nodal concentration
 - Cause of the disease is unknown but appear to involve either the body's production of auto-antibodies that react with myelin basic protein or the secretion of proteases that destroy myelin proteins
- Onset btw 20-40 years; Women more than men
- Sclerotic plaques or lesions around demyelinated neurons (via MRI)
- Spasms and weakness in one or more limb; ataxia; bladder dysfunction; local sensory losses; visual disturbance; slurred speech; loss of coordination
 - Others: fatigue; cognitive impairment; depression; unstable mood; nystagmus; optic neuritis; diplopia; dysarthria; dysphagia; pain; hypoesthesia; paraesthesia; diarrhea or constipation

- Tx (immune suppressor agents)
 - Interferon β
 - Modulates the responsiveness of the immune system
 - Slows the progression and severity of attacks. Prevents relapse
 - Corticosteroids (during active periods)
 - Glatiramer acetate (copaxone)
 - Peptide composed of random sequences of glutamate, alanine and lysine
 - May stimulate anti-inflammatory response

Myasthenia Gravis

- Immune system attack and **destroys acetylcholine receptors** at the neuromuscular junction → Muscle weakness
- Ocular myasthenia gravis → restricted primarily to ocular-motor system
- Cause is unknown but appears to be an autoimmune disorder involving the **thymus**
- 20-40 years; 12% of infants born to women with this disease have a syndrome called neonatal myasthenia due to antibodies passively crossing placenta (lasts only a few weeks); It is so rare, it is an orphan disease
- Symptoms: Ptosis; Diplopia (double vision); Muscle fatigue after exercise
 - Other indicator: Ab for Ach receptor are present in the serum
 - 50% in ocular form; 90% in generalized form
- Dx: Short-acting cholinesterase inhibitor (edrophonium) to reverse presenting symptoms
 - Patient shows immediate and short-lived improvement
- Tx
 - Cholinesterase inhibitors (manage the symptoms)
 - Corticosteroids and immunosuppressive drugs (may retard autoimmune response)
 - Thymectomy (removal of thymus/helpful in patients with generalized form)

CNS INFECTION

- Most meningeal infections involve the leptomeninges
- Life threatening = Medical emergencies
- Cortex and the spinal cord are found in confined spaces= cant tolerate the swelling due to inflammation
- Inflammation and edema=> tissue infarction => permanent neurologic sequel or death.
- BBB => difficult to cross =>protects from pathogens, toxic substances and serum components of the immune system =>delay immune response to an infection in the CNS
- Routes of infection:- hematogenous spread(most common)=> arterial route or retrogradely (veins), direct implantation (traumatic or iatrogenic via lumbar puncture), local extension and PNS into CNS (rabies, herpes zoster, tetanus , poliomyelitis)

Acute bacterial meningitis

- ☐ symptoms hours-several days
- ☐ Bacterial infection involving the leptomeninges
- ☐ almost always a bacterial infection
- ☐ from:- Skull or backbone fractures, Medical procedures, Along peripheral nerves and Blood or lymph
- ☐ via blood from cardiac, pulmonary, GIT, sinus, ear, nasopharyngeal infections
- ☐ via blood abscess from otitis media and sinus infection
- ☐ via cribriform plate defect from nasopharynx
- ☐ cause:-age, immune states and whether the infection is community acquired or nosocomial
- neonates=>E.coli, S.agalactiae and listeria monocytogene
- 1mth-50yrs=>S.pneumonia(community acquired) and N.meningitidis(crowded areas)
- >50yrs=> S.pneumonia and listeria monocytogene
- Nosocomial=> gram negative(e.coli), following endocarditis(s, aureus), following ventricular shunt replacement(s.epidermidis, s.aureus, enterococcus, bacillus subtilis, corynebacterium)
- ☐ 3 phases- Colonization by the pathogen, Invasion of the CNS and host's response to infection
- Colonization - nasopharyngeal colonization
- VF- polysaccharide capsules, pili, and enzymes => adherence of organisms to nonciliated

nasopharyngeal epithelium and establish mucosal colonization. IgA1 protects the URT by inhibiting the adherence of bacteria and preventing penetration into mucosal tissue.

– Invasion of the CNS=>Most common via bloodstream from the ears, sinuses, throat, lungs, heart, and GIT. From blood it can settle in large venous sinuses then penetrate the dura and arachnoid and infract the CSF.

☐ Less common:- breaks in the cribriform plate or defects in the base of the skull following basilar skull fractures

☐ Trauma=>leakage of CSF into the middle ear or sinuses =>gain entrance into the subarachnoid space=>bacteria multiply easily

– The host's response :- bacterial growth attracts PMNs to infection site

☐ The PMNs lyse=> release toxic factor =>necrosis and edema

☐ Marked inflammatory response => negative consequences=>BBB becomes more permeable to serum protein =>decreased glucose entrance to arachnoid and pia matter=>increased protein and decreased glucose in the CSF

☐ Inflammation=>decrease CSF flow=> cerebral edema

☐ Has become a disease of adults and is relatively uncommon.

☐ S .pneumoniae is spread person to person by respiratory droplets, and is the most common cause.

☐ Listeria monocytogenes acquired following ingestion of commonly contaminated foods such as dairy products.

☐ L monocytogenes meningitis occurs more often in patients with depressed cellular-mediated immune responses.

☐ an upper respiratory tract infection (e.g., sore throat, rhinorrhea, and nasal congestion) or ear infection may precede.

☐ an abrupt onset of worsening fever and one or more meningeal symptoms.

☐ The most common manifestations of acute bacterial meningitis include: headache, fever, nuchal rigidity, and altered mental status (neurologic findings)

☐ Evidence of otitis media, Pharyngeal inflammation.

☐ Purulent nasal discharge is an indication that the patient may have a sinusitis infection that has resulted in meningitis.

☐ If the patient has pneumonia, the acute bacterial meningitis that results is usually due to S pneumoniae.

☐ *Neisseria meningitidis* and *Streptococcus pneumoniae* account for 37% to 93% of acute bacterial meningitis cases:-

☐ Fever- 95% of individuals and can last 4-8 days after therapy

☐ Generalized severe unremitting headache - radiate down the neck and anti pains do not relieve pain

☐ Neck stiffness(nuchal rigidity)- 90% of patients, Brudzinski or Kernig sign

☐ Altered mental status- in 80% of patients lethargic, stuporous, difficult to arouse, confused, disoriented, or in a coma. More severe cases result in loss of consciousness and grand mal or focal seizures.

☐ CAN PROGRESS IN A FEW HOURS ESPECIALLY MENINGOCOCCAL MENINGITIS AND PNEUMOCOCCAL MENINGITIS.

☐ MINUTE(1-2mm) HEMORRHAGES INTO THE SKIN OR MUCOSAL MEMBRANES(PETECHIAE) OR LARGER(1-2cm) SUBCUTANEOUS HEMATOMAS(ECCHYMOSES) ARE IMPORTANT CLUES IN THE DIAGNOSIS OF MENINGOCOCCAL DISEASE.

☐ IF SEVERE HEADACHES ARE PRESENT IT IS MOST LIKELY A BACTERIAL MENINGITIS THAN VIRAL.

Aseptic meningitis (Viral Meningitis)

☐ most common type of meningitis and usually the result of a viral infection

☐ Clinical course is less fulminant compared to bacterial

☐ enterovirus subgroup of picornaviruses are responsible for at least half of the cases of viral meningitis.

☐ Echoviruses and coxsackieviruses A and B: are the most common causes of aseptic meningitis

☐ Human simplex virus type 2 (HSV-2): aseptic meningitis can occur during primary genital infection

☐ Varicella-zoster virus (VZV), Epstein-Barr virus (EBV), and cytomegalovirus (CMV)

☐ Human immunodeficiency virus (HIV) aseptic meningitis occurs during primary infection in 5–10% in patients who are HIV positive

☐ Enteroviruses spread via direct contact with an infected respiratory secretion and feces.

☐ Mumps, herpes, chicken pox are spread through personal contact.

☐ Echoviruses and coxsackieviruses gain spread through blood.

☐ HSV-1, HSV-2, VZV through the neural route from the nerve roots.

- ☐ The viruses initially replicate in their target organ system (respiratory or GIT mucosa) and then gain access to the bloodstream.
- ☐ The primary viremia seeds the virus in the reticuloendothelial organs
- ☐ If the replication continues and a secondary viremia occurs, the viruses then can infect the CNS
- ☐ Cross the BBB=> directly at the capillary endothelial level or via natural defects
- ☐ Inflammatory response is seen in the form of pleocytosis [increase in WBC]
- ☐ Polymorphonuclear (PMNs) leukocytes are most abundant within the first 24–48 hours followed later by increasing numbers of monocytes and lymphocytes.
- ☐ The CSF of a patient with viral meningitis will contain: - high protein, normal glucose concentration, and high leukocyte count of primarily lymphocytes.
- ☐ The course of viral meningitis is slower and more indolent than that of bacterial meningitis
- ☐ Arboviruses and enteroviruses are endemic in the summer and early fall.
- ☐ Think acute HIV infection with aseptic meningitis, especially if rash, pharyngitis, and lymphadenopathy are present.
- ☐
- ☐ Infant meningitis
- ☐ Viral, bacterial or parasitic
- ☐ Bacterial causes are the most common cause.
- ☐ Acquired within one month after birth.
- ☐ If diagnosed after one month viruses tend to be more common cause of meningitis.
- ☐ Causes:-
- ☐ Enteroviruses-most common for children <3 mth
- ☐ HSV-vesicular skin lesion may be in clusters
- ☐ VZV-vesicular rash 9-15 days after appearance of mother's rash, the lesion appears all over the body
- ☐ CMV
- ☐ E.coli and S.agalactiae- most common cause for children <1 mth
- ☐ Listeria monocytogenes- occasional

- ☐ *Toxoplasma gondii*
- ☐ When a neonate or infant presents with fever and lethargy or irritability, it is important to consider the diagnosis of meningitis even if the classic signs and symptoms are absent

(i.e., fever, nuchal rigidity, and change in mental status).

- ☐ Other symptoms could be:-
 - Poor feeding, vomiting, paradoxical irritability
 - Respiratory distress, apnea, and cyanosis
 - Bulging fontanelle
 - Hypotonia

- ☐ Chronic meningitis

- ☐ Most common cause of the it is *M.tuberculosis* in immunocompromised patients.

- ☐ *Cryptococcus neoformans* is the most common cause of chronic meningitis in AIDS patients, and is the second most common infectious cause of chronic meningitis in all patients regardless of immune status.

- ☐ Chronic meningitis due to *M. tuberculosis* results from rupture of a tubercle into the adjacent subarachnoid space.

- ☐ *C .neoformans* produces a capsule and melanin that protect the yeast cells from being phagocytized

- ☐ CRYPTOCOCCAL MENINGOENCEPHALITIS

- ☐ By the encapsulated yeast, *Cryptococcus neoformans*.

- ☐ Uncommon disease until the onset of the AIDS epidemic.

- ☐ Most common fungal cause of chronic meningitis is *Cryptococcus neoformans*, most often in patients with HIV/AIDS.

- ☐ The organism is especially abundant in soil containing bird droppings

- ☐ *C. neoformans* is an oval, budding yeast surrounded by a wide polysaccharide capsule

It is not dimorphic.

- ☐ Forms a narrow-based bud, whereas the yeast form of *Blastomyces dermatitidis* forms a broad-based bud.

- ☐ *C. neoformans* is airborne and enter through inhalation.

- ☐ In the lung symptoms could be mild or no symptoms.

- ☐ IC patients=> organisms multiply and enter bloodstream and distributed throughout the body.
 - ☐ The capsule consisting of negatively charged polysaccharides that are immunosuppressive => blocking both cell-mediated immune responses and leukocyte migration.
- => These effects explain the minimal inflammatory response elicited by invading
- ☐ Meningoencephalitis cause thickening of the meninges=>impeding the flow of CSF=>increase pressure.
 - ☐ Humans are the natural host for polio viruses, coxsackieviruses, and echoviruses.
 - ☐ Person to person by direct or indirect fecal–oral contact.
 - ☐ May occur by respiratory route=>URT infection
 - ☐ Enteroviruses=> young children are most susceptible.
 - ☐ Viral meningitis is 5–8 times more common in infants than in adults. And children has higher morbidity and mortality rates.
 - ☐ The non-polio enteroviruses (i.e., echovirus, coxsackieviruses A and B viruses) are the most common cause of aseptic meningitis.
 - ☐ Enteroviruses cause approximately 85% of all cases of meningitis
 - ☐ Coxsackievirus B causes over 60% of meningitis cases in children younger than 3 months of age.
 - ☐ HSV-1, HSV-2, CMV, and HIV-1 can be transmitted via sexual contact.
 - ☐ VZV is transmitted by respiratory droplets.
 - ☐ EBV, HSV-1, and CMV are transmitted via infected saliva and other mucous secretions and are spread person to person.
 - ☐ fever, headache, and nuchal rigidity
 - ☐ quite similar to those observed in acute bacterial meningitis
 - ☐ Unique manifestations include:- maculopapular rashes and conjunctivitis (an echovirus) and lethargy but with clear mentation and no focal neurologic findings.

DIAGNOSIS

- ☐ To obtain a definitive diagnosis of meningitis, it is necessary to perform a lumbar puncture
- ☐ After lumbar puncture, empiric antimicrobial therapy can be initiated in most cases. The highest survival rates in cases of acute bacterial meningitis are obtained when antibiotics are administered to the patient within 30 minutes of arrival in the emergency department.

- ☐ A minimum of at least 1mL of CSF should be placed into each of the four sterile tubes
- =>One tube will be used for bacterial Gram stain and culture, One for cell count and differential, One for CSF glucose and protein analysis, and The fourth one should be used for viral culture or saved at -70°C for future use.
- ☐ Concentrating the CSF before Gram staining increases the ability to detect 1000–10,000 bacterial cells/mL of CSF.
- ☐ If no antibiotics are given before the LP, the sensitivity of the Gram stain is about 92%.
- ☐ Gram stain of the petechial lesions as well as a cytocentrifuged Gram stain of CSF revealed many WBCs and moderate intracellular and extracellular gram-negative diplococci
- ☐ Media used routinely when performing bacterial cultures of CSF include :- 5% sheep blood agar, chocolate agar and an enrichment broth to detect small numbers of bacteria.
- ☐ If the bacteria are growing on both blood and chocolate agars, the bacteria are not *H. influenzae*.
- ☐ CSF can be analyzed for *M. tuberculosis* by acid-fast staining.
- ☐ *Cryptococcus* is the only encapsulated yeast that causes human infection and can be identified by an India ink stain of the CSF.

CSF FINDINGS

- ☐ ASEPTIC MENINGITIS
- ☐ Opening pressure- normal or slightly elevated
- ☐ Clear or slightly turbid
- ☐ Increased protein
- ☐ Normal or slightly decreased glucose
- ☐ Increased WBC count=> lymphocytes
- ☐ No microorganisms observed after stain or culture
- ☐ CSF:serum glucose ratio=>>0.6
- ☐ ACUTE BACTERIAL MENINGITIS
- ☐ Opening pressure elevated
- ☐ Turbid may even clot
- ☐ Increased protein
- ☐ Decreased glucose
- ☐ CSF:serum glucose ratio=>< 0.3

- ☐ Increased WBC count=>PMNs predominate
- ☐ Show organisms in 75% untreated cases
- ☐ Organisms grow in culture in 85% of untreated cases
- ☐ CHRONIC(TUBERCULOUS AND FUNGAL) MENINGITIS
- ☐ Normal or mildly increased for TB and increased in fungal
- ☐ Clear to slightly turbid, if left to stand a pellicle or slightly visible "spider's web clot"
- ☐ Increased protein(224mg/dL)
- ☐ Decreased glucose
- ☐ CSF:serum glucose ratio=><0.4
- ☐ Increased PMNs in TB and increased lymphocytes in fungal
- ☐ 25-40% positive acid fast for TB and 80-90% of fungal positive for indian ink.
- ☐ TB culture can take up to 4 weeks

TETANUS

- ☐ Affects primarily the nerves of the spinal cord.
- ☐ Tetanus toxin prevents release of these inhibitory neurotransmitters and causes the muscles to remain in spasm
- ☐ Tetanus is caused by the anaerobic spore-forming, gram positive, rod-shaped bacterium *C. tetani*.
- ☐ General characteristics of clostridium
- ☐ Possess Endospores, Most strains are strict anaerobic and some are aerotolerant, Are Gram positive bacilli, Ubiquitous in soil, water, sewage, part of normal flora of intestine of animals and humans and Most are harmless /saprophytes, Proteolytic
- ☐ Large, motile, Produce round, terminal spores that gives appearance of a drumstick
- ☐ Causes tetanus or lockjaw, a neuromuscular disease
- ☐ Spores of some strain resistance boiling in water for up to 3 hours
- ☐ After entering into the host the spore germinates to vegetative stage which can produce toxin in our body
- ☐ It produces two toxins:- an oxygen labile hemolysin (tetanolysin) and heat labile neurotoxin called tetanospasmin
- ☐ Tetanolysin is a heat stable hemolysin of unknown significance.

☐ Tetanospasmin initially binds to peripheral nerve terminals=> The toxin is then transported within the axon and across synaptic junctions until it reaches the CNS=> Once in the CNS, it binds to gangliosides at the presynaptic inhibitory motor nerve endings and is taken up into the axon=> The toxin blocks the release of inhibitory neurotransmitters=> Tetanospasmin cleaves a protein component of synaptic vesicles called synaptobrevin II, which aids in release of the inhibitory neurotransmitters into the synaptic cleft

☐ If no inhibitory NT can be released, the muscle will remain in spasm, causing the generalized muscular spasms characteristic of tetanus.

☐ The shortest PN are the first to deliver the toxin to the CNS, which leads to the early symptoms of facial distortion and back and neck stiffness.

☐ The spores enter the body through breaks in the skin and germinate under low oxygen conditions.

☐ Puncture wounds and wounds with a significant amount of tissue injury are more likely to promote germination.

☐ Common in areas where soil is cultivated, in rural areas, in warm climates

☐ Tetanus is NOT transmitted person to person.

☐ IP- 2-21 DAYS

☐ The further the injury the longer the IP

☐ 3 types of Tetanus

1. Local tetanus

☐ Is an uncommon form of the disease in which patients have persistent contraction of muscles in the same anatomic area as the injury. The contraction gradually subsides.

☐ Precede the onset of generalized tetanus but is generally milder.

2. Cephalic tetanus

☐ Is a rare form of the disease, occasionally occurring in conjunction with otitis media or following injuries to the head.

☐ There is involvement of the cranial nerves, especially in the facial area

3. Generalized tetanus

☐ Is the most common type (about 80%) of tetanus.

☐ The spastic paralysis usually presents with a descending pattern=>The first sign is trismus (lockjaw), followed by stiffness of the neck, difficulty in swallowing, and board-like rigidity of the abdominal muscles.

- ☐ Spasm of the facial muscles leads to a grotesque grinning expression termed risus sardonicus.
- ☐ Spasm of the somatic musculature may result in opisthotonos, a form of spasm in which the head and heels are bent backward and the body bowed forward
- ☐ When spasms occur, they may be frequent and last for several minutes. Spasms continue for 3–4 weeks.
- ☐ Other symptoms include a temperature rise of 2–4 °C above normal, sweating, elevated blood pressure, and episodic rapid heart rate.

Neonatal tetanus

- ☐ generalized tetanus that occurs in newborn infants
- ☐ Symptoms usually appear 4–14 days after birth.
- ☐ Neonatal tetanus occurs in infants born without protective passive immunity because the mother is not immune.
- ☐ It usually occurs through infection of the unhealed umbilical stump, particularly when the stump is cut with contaminated instruments.

DIAGNOSIS

- ☐ There are no laboratory findings characteristic of tetanus.
- ☐ The diagnosis is determined based on clinical presentation of the disease.
- ☐ Early clues to the diagnosis of tetanus include a recent wound, no clear history of tetanus immunization, irritability, restlessness, headache, and low-grade fever

Treatment and prevention

- ☐ Treatment of tetanus are not satisfactory
- ☐ Prevention is all-important

=> Active immunization with toxoids: DPT (diphtheria, pertussis, tetanus), tetanus toxoid, antigenic, no exotoxic activity

=> Proper care of wounds contaminated with soil

=> Prophylactic use of antitoxin

=> Administration of penicillin

- ☐ IM administration 250–500 units of human antitoxin (tetanus immune globulin).
- ☐ Patients with tetanus should receive muscle relaxants, sedation, and assisted ventilation
- ☐ Penicillin strongly inhibits the growth of *C. tetani* and stops further toxin production.

POLIO (Polio virus)

- ☐ Poliomyelitis (POLIO) acute infectious disease affecting CNS =>Destruction of motor neuron in the spinal cord resulting in flaccid Paralysis.
- ☐ Virus member of Picornaviridae family
- ☐ Picorna viruses are very large (in number) but one of the smallest sized RNA viruses(28-30 nm)
- ☐ Most diverse (more than 200 serotypes) and 'oldest' known viruses-1400 BC
- ☐ Clinically classified as Enteroviruses and Rhinoviruses
- ☐ Clinically significant :-Poliovirus, Hepatitis-A virus and Human Rhinoviruses
- ☐ Non enveloped, Icosahedral symmetry
- ☐ Genome : ss RNA, linear, positive sense
- ☐ Four capsid proteins VP1-AP4 from cleavage of the original protomer VP0 +single genome linked protein (VPg)
- ☐ One or two nonstructural proteins
- ☐ Replication and assembly in cytoplasm
- ☐ Infection: acute and cytolytic / persistent
- ☐ Insensitive to chloroform, nonionic detergents and sodium desoxycholate
- ☐ Enteroviruses are stable at PH of 3-5, Rhinoviruses are acid labile.

Enteroviruses

- ☐ Humans are the only known reservoir
- ☐ Stools →sewage → contamination of water supplies can travel long distances
- ☐ Contamination of sea food
- ☐ survive exposure to the sewage treatments and chlorination

polio viruse

- ☐ is a typical enterovirus
- ☐ Three serotypes: 1, 2, 3=>Minimal heterotypic immunity between serotypes
- ☐ Rapidly inactivated by heat, formaldehyde, chlorine(much higher concentrations needed), ultraviolet light
- ☐ Narrowest tissue tropism (anterior horn cells of spinal cord, motor neurons, skeletal muscle cells, lymphoid cells)

- ☐ Entry through mouth=>Replication in pharynx, GI and local lymphatic
- ☐ Hematologic spread to lymphatic and CNS=>Viral spread along nerve fibers=>Destruction of motor neurons
- ☐ IP- 3 - 35 days, Usually 7-14 days
- ☐ Asymptomatic infection, to mild fever, to severe and permanent paralysis
- ☐ 1/200 - irreversible paralysis (usually in the legs).
- ☐ Between 5-10 % infected die when their breathing muscles are paralyzed.
- ☐ Polio mainly affects children under five years of age.
- ☐ No cure for polio thus, the best treatment is preventive. A few drops vaccine will protect a child for life.

1. Abortive poliomyelitis-most common

- Minor illness
- Fever, malaise, drowsiness, head ache, nausea, Vomiting, constipation, sore throat
- Recovery in few days

2. Non paralytic poliomyelitis-aseptic meningitis

- Symptoms As with type 1
- Stiffness and pain in back and neck
- Disease 2-10 days → recovery complete

3. Paralytic poliomyelitis

- Lower motor neuron damage → Flaccid paralysis
- Maximal recovery 6 months
- Residual paralysis much longer

4. Progressive Post poliomyelitis(0.5-2 % cases)

- Most severe form Virus spreads from blood to anterior horn cells of spinal cord and motor cortex of brain
- Spinal paralysis of one or more limbs
- Permanent paralysis and muscle wasting

Diagnosis

- ☐ Recovery of virus from throat swab/rectal swab

- ☐ Specimen frozen during transit
- ☐ Human / monkey cells => culture
- ☐ Nt
- ☐ CPE 3-6 days
- ☐ Serology: Very rarely used for diagnosis since cell culture is efficient Paired serum – Nt

Epidemiology

- ☐ Human is only reservoir
- ☐ Feco-oral or oral-oral transmission
- ☐ 7-10 days before onset Virus present in stool
- ☐ Passive immunity mother to offspring- maternal Ab → disappears in 6 months

Killed Polio Vaccine (Salk Vaccine)

- Contains 3 serotypes of vaccine virus
- Grown on monkey kidney cells
- Inactivated with formaldehyde
- 4 inoculations over a period of 1-2 yrs
- Periodic boosters
- Only IgG
- >90% immune after 2 doses
- >99% immune after 3 doses
- Duration of immunity not known with certainty
- Can be given In Pregnancy

Live Oral Polio Vaccine (Sabin)

- Contains 3 serotypes of vaccine virus
- Grown on monkey kidney or human diploid cells
- Stabilized with Mag Chloride
- Shed in stool for up to 6 weeks following vaccination
- Problems: Storage, Mutation (type 2 and 3 mostly) Interference with other enteroviruses
- IgG, IgA and IgM

- ☐ Pediarix-1 vaccine- Contains IPV, DTaP, and hepatitis B vaccines(diphtheria, tetanus toxoids and pertussis, and hepatitis B vaccines and inactivated polio vaccine)

Slow viral infections

1. Subacute sclerosing panencephalitis SSPE (Dawson Disease)

- ☐ Persistent infection of brain with Measles
- ☐ IP may be ` 20 years
- ☐ Rare disease in young adults
- ☐ Slow progressive demyelination in CNS =>death(within 3 years)
- ☐ No cure
- ☐ 1st stage-behavior becomes more abnormal and erratic, memory loss and mental deterioration
- ☐ 2nd stage-intensity of the spasms and the mental deterioration increases. loss of the ability to walk, speech impairment and coma

2. Progressive Multifocal Leukoencephalopathy

- ☐ JC virus- Papovaviridae family
- ☐ Progressive multifocal areas of demyelination scattered throughout CNS
- ☐ Immunocompromised / AIDS patients
- ☐ Confusion, personality changes, visual deficits, mental impairment, motor weakness
- ☐ Death is invariable & occurs within weeks to months of clinical onset.

Borna Disease

- ☐ CNS disease primarily of horses and sheep in certain areas of Germany
- ☐ Humans can be infected
- ☐ Persistent infections
- ☐ Neuropsychiatric disorders
- ☐ 90 nm, Enveloped ss neg sensed RNA virus-replication nuclear =>Bornaviridae

Progressive Rubella Panencephalitis

- ☐ Neurologic disorder which may occur in a child with congenital rubella- very rare
- ☐ slow viral infection of the brain characterized by chronic encephalitis
- ☐ Age between 8–19 years of age

- ☐ Reactivation of persistent rubella virus infection
- ☐ Severe neurologic deterioration=>invariably fatal

PRION

- ☐ Small proteinaceous infectious particle
- ☐ No traceable nucleic acid
- ☐ Prion protein PrP sialoglycoprotein –with MW 33,000 to 35,000
- ☐ Normal form of protein PrP^c – Abnormal form PrP^{sc} Transmissible protein
- ☐ No antibiotics can cure disease caused by prions
- ☐ They are not typical of a prokaryotic organism or a eukaryotic organism or a virus
- ☐ All that is present in this pathogen is the protein PrP^{Sc}, the mutation of PrP^c
- ☐ PrP^{Sc} is resistant to any form of digestion
- ☐ Prions are non immunogens
- ☐ They are resistant to high temperatures & disinfectants
- ☐ PRIONS cause spongiform encephalopathies which are slow degenerative diseases of CNS

☐ HUMAN DISEASES:- Kuru, Creutzfeldt-Jacob disease (CJD), Gerstmann-Straussler-Scheinker (GSS) disease and Fatal familial Insomnia (FFI)

☐ ANIMAL DISEASES:- Scrapie, Bovine spongiform encephalopathy (BSE) (mad cow disease), Chronic wasting disease and Transmissible mink encephalopathy

☐ Diagnosis (autopsy) – brain cytopathology

☐ Treatment: none; uniformly fatal

☐ MAD COW DISEASE

☐ The Mad Cow Disease in Cows, Scrapie in Sheep, the Creutzfeldt-Jakob Disease in human

☐ Initially thought to be due to “slow viruses” because of their long IP these are now known to be caused by agents called prions.

☐ Source cattle feed contaminated with bone meal from scrapie infected sheep / BSE infected cattle carcasses

☐ KURU

☐ Sub acute progressive neurological disease, Human spongiform encephalopathy

- ☐ Only eastern highlands of New Guinea
- ☐ Cannibalism
- ☐ IP about 40 months
- ☐ Ataxia, disturbed balance, clumsy gait, tremors → death 5-12 months
- ☐ CREUTZFELDT-JACOB DISEASE (CJD)
- ☐ Degenerative neurologic disorder that is incurable and invariably fatal.
- ☐ Also called a human form of mad cow disease though, not related to BSE
- ☐ ages 55 – 65 Years
- ☐ Diagnosis: CSF 2 proteins, 130,131
- ☐ Histologic study of brain material-spongiform changes
- ☐ No treatment
- ☐ Variant Creutzfeldt-Jakob disease
- ☐ Rare and fatal human neurodegenerative condition.
- ☐ Consumption of food Bovine Spongiform Encephalopathy
- ☐ In contrast to the traditional forms of CJD, vCJD has affected younger patients (median age at death of 28 years, as opposed to 68 years) and has a relatively longer duration of illness (median of 14 months as opposed to 4.5 months).
- ☐ Gerstmann-Straussler-Scheinker syndrome
- ☐ Dementia with progressive cerebellar degeneration
- ☐ Mean duration to death is 5 years.
- ☐ Most cases of G-S-S disease have been familial in an autosomal dominant pattern but tissue is able to transmit disease to animals.
- ☐ This appears to be an allelic variant of familial Creutzfeldt-Jakob disease.
- ☐ SCRAPIE
- ☐ Sheep and goats
- ☐ Transmission to mice and hamsters, no transmission to humans
- ☐ Infection: brain, spinal cord, and eye
- ☐ Neurologic symptoms development of amyloid plaques in CNS – PrPsc level high in brain

Toxoplasma gondii

- Intracellular Protozoan of the coccidian family
- Felines are the definitive host
- Ingestion of oocyst/cyst, Sporozoites=>Invade nucleated cells and differentiate into tachyzoites=>Based on host immune response- it divided and distributed rapidly- (CNS, lungs and eye)- rarely symptomatic
- Tachyzoites differentiated into latent bradyzoite- because of immune reconstitution-leads to chronic stage of infection
- Surrounded by a CHO-rich cyst wall within the parasitophorous vacuole
- Responsible for encephalitis or chorioretinitis

Immunity and toxoplasmosis

- Th-1 initiation of CMI =>Astrocytes, microglia and APCs produce IFN- μ and activate CD4+T cells
- CD4+T cells produce IL-2, IL-12, that
 - => Facilitate phagocytosis (Macrophages, DCs)
 - =>Opsonization, activation of NK cells (NO sources)
 - =>Cytotoxic T cells (CD8+T cells)
- Th-2- humoral immunity- Ab production by B cells
- Alteration of protein synthesis is highly associated for toxoplasmosis
- Acetylation alter the function, location and other aspects of proteins in Astrocytes, and causes 'Rewiring' of information and affect behavioral change in rodents and humans
- In IC patient If CD4+ cells are lowered than 200 CD4 T cells/ μ L there will be reduced activity of cytotoxic T cells, NK cells and IFN-g=> reactivation of bradyzoites
- If CD4+ cells are less than 100 / μ L=> clinical symptoms start
- Organ dependent: CNS (80%), Retina (5-10%), Pneumonitis, Myocarditis,
- Fever and headache (40-70%), Focal neurologic signs (50-60%),
- Speech disturbance, movement disorders and neuropsychiatric symptoms
- Cranial nerve palsies, Seizures (30-40%),
- Diffuse neurologic dysfunction including, confusion and lethargy (40%), unconsciousness and death

Diagnosis

- ☐ Serological- mostly IgG through ELISA. Sometimes comparative test of CSF IgG titration with serum
- ☐ Imaging's like CT, MRI, or other radiographic testing
- ☐ Detection of the organism in a clinical sample:- Brain biopsy-via stereotaxic CT method=> treatment non-respondents /failed patients
- ☐ Lumbar puncture for:- PCR or Other cytology culture

Prevention and treatment

1. Preventive exposure
2. Primary Prophylaxis:- Trimethoprim-sulfamethoxazole (TMP-SMX) or Dapsone-pyrimethamine plus leucovorin (if they are TMP-SMX intolerance)
3. Treating Disease (if symptomatic):- Pyrimethamine(penetrates brain) plus sulfadiazine plus leucovorin(prevent hematologic toxicity)
4. Preventing Recurrence (Chronic maintenance therapy)

Severity

- ☐ Mode of Infection:-Tachyzoite phase is highly symptomatic than others
- ☐ Parasite strain types:- Human infection is common by T.gondii-type-II
- ☐ Host cytokine response:- IFN-gamma-dependent, cell-mediated immune responses, NK cells, CD8+T cells and others

EXIT EXAM NOTE ON PATHOLOGY

CNS TUMOR

GLIOMAS

- include *astrocytomas*, *oligodendrogliomas*, and *ependymomas*.
- **ASTROCYTOMA**
 - **Pilocytic Astrocytoma**
 - Circumscribed, well-differentiated tumor of childhood that may occur in many CNS locations but usually presents as a cystic cerebellar mass. Mitoses are rare.
 - biphasic appearance—dense, coarsely fibrillar areas and loose, microcystic areas.
 - WHO Grade I
 - Most common tumor of childhood.
 - Symptoms depend on site of neoplasm: Tumors obstructing fourth ventricle or aqueduct can cause hydrocephalus and symptoms of headache, nausea, vomiting. Optic nerve tumors may cause vision loss.
 - Prognosis is typically good, with 20-year survival in approximately 80%.

- Lesions associated with neurofibromatosis type 1 (NF1) have a better prognosis
- Diffuse astrocytoma
 - Infiltrating glial neoplasm with astrocytic features, typically with some degree of pleomorphism, increased cellularity and slow growth.
 - World Health Organization (WHO) Grade II.
 - Peak incidence in fourth decade, with wide age range.
 - Presentation: Typically found within the cerebral hemispheres, with propensity for frontal and temporal lobes, but can be found anywhere in the central nervous system.
 - Pathology: Ill-defined lesions with blurring of the gray-white matter junction. May be gray or yellow-white without tissue destruction. Mitotic activity absent or very rare; no necrosis or microvascular proliferation.
 - Symptoms: Seizure, speech difficulty, behavior changes, or motor deficits.
 - Treatment: Treatment includes surgical resection and radiotherapy.
 - Typically progress toward a higher-grade astrocytic glioma.
- Anaplastic astrocytomas
 - Intermediate-grade infiltrating glioma derived from malignant astrocyte-like cells, typically with increased cellularity, nuclear atypia, and mitotic activity; may arise from lower-grade diffuse astrocytoma, and may progress to glioblastoma.
 - WHO Grade III
 - Peak incidence occurs in fifth decade of life.
 - Seizures and focal neurologic deficits are common.
 - Ill-defined tumor that infiltrates brain without associated tissue destruction.
 - Can produce a more discernable tumor mass compared to diffuse astrocytoma.
- Glioblastoma
 - The most malignant glioma—predominant astrocytic differentiation with nuclear and cellular pleomorphism, mitotic activity, microvascular proliferation, thrombosis, and necrosis.
 - Poor prognosis, average survival less than 1 year.
 - Poorly defined lesion with peripheral area of grayish tumor and central areas of yellow-tan necrosis.
- Oligodendroglioma
 - A diffuse infiltrating glioma of adults consisting of cells resembling oligodendroglia,
 - Because of cortical location, two thirds of patients present with chronic seizures.
 - Soft pink-gray lesion that is ill-defined with blurring of gray-white junction
 - Calcifications may be present, giving a gritty consistency to the tumor.
 - Monomorphic tumor cells with tendency to form small clusters.
 - individuals with oligodendrogliomas have a better prognosis than do those with astrocytomas.
 - **Anaplastic oligodendrogliomas:** grade III/IV) are characterized by increased cell density, nuclear anaplasia, increased mitotic activity, and necrosis.

- **NEURONAL TUMORS**

- Ganglion cell tumor
 - Well-differentiated tumors of the CNS with mature neurons (ganglion cells) as a defining feature: includes ganglioglioma (ganglion cell tumor with a low-grade glial element), gangliocytoma (without a glial element), and anaplastic ganglioglioma (ganglion cells and anaplastic glial elements).
 - Associated with seizures; location-dependent symptoms.
 - Most are benign (WHO Grade I)
 - Demarcated or discrete tumor mass; with or without cystic changes or calcifications.
 - Ganglion cells may exhibit binucleation or have bizarre-appearing nuclei.

Ependymomas

- typically slow-growing neoplasms of children and young adults that originate from the ependymal lining of the cerebral ventricles or central canal of the spinal cord.
- Circulate through CSF

- **EMBRYONAL (PRIMITIVE) NEUROEPITHELIAL TUMORS**

- Medulloblastoma
 - Malignant, embryonal tumor arising primarily in the posterior fossa of children.
 - Most common malignant brain tumor of childhood.
 - Solid masses of friable gray-white tissue, apparently Circumscribed Usually involves surface of cerebellar folia and infiltrates leptomeninges.
 - WHO Grade IV
 - Dissemination through the CSF is a common complication, presenting as nodular masses elsewhere in the CNS, including metastases to the cauda equina that are sometimes termed drop metastases.

- **PRIMARY CNS LYMPHOMA (PCNSL)**

- Malignant lymphomas arising independently within the CNS without evidence of systemic lymphoma; most PCNSLs are aggressive B-cell lymphomas.
- Individuals with inherited or acquired immunodeficiency are predisposed to PCNSL.
- Strong link to Epstein-Barr virus in immunocompromised patients with PCNSL.
- Focal or multifocal neurologic deficits, neuropsychiatric symptoms, symptoms of elevated intracranial pressure, seizures
- primary lymphomas of the CNS are aggressive, with relatively poor response to chemotherapy compared with peripheral lymphomas.

- Intravascular “angiotropic” large B-cell lymphoma

- Neoplastic B cells accumulate within and obstruct vasculature causing multiple microinfarcts
- More than 30% of reported cases have neuraxis involvement
- Multiple organ systems can also be affected.

- **MENINGIOMAS**

- Neoplasms arising from meningotheial (arachnoidal) cells; are usually attached to the inner surface of the dura mater (dural-based tumors).
- Well-demarcated, firm, rubbery, yellow-tan, round to lobulated mass attached to dura

- Usually compresses adjacent brain or spinal cord
- May invade skull and cause hyperostosis (thickening of the cortical bone).
- Most meningiomas are WHO Grade I and are generally treated by gross total resection; may recur if not complete excised.
- **METASTATIC TUMORS**
 - The five most common primary sites are lung, breast, skin (melanoma), kidney, and gastrointestinal tract, accounting for about 80% of all metastases.
 - Some rare tumors (e.g., choriocarcinoma) have a high likelihood of metastasizing to the brain, whereas some more common tumors (e.g., prostatic carcinoma) almost never grow in the brain.

EYE

Orbit

PROPTOSIS

- Diseases that increase orbital contents therefore displace the eye forward.
- proptotic eye might not be covered completely by the eyelids, and the tear film might not be distributed evenly across the cornea. Corneal exposure is painful and can predispose to corneal ulceration and infection.

THYROID OPHTHALMOPATHY (GRAVES DISEASE)

- Proptosis is caused by the accumulation of extracellular matrix proteins and variable degrees of fibrosis in the rectus muscles.

NEOPLASMS

- most frequently encountered primary neoplasms of the orbit are vascular in origin: the capillary hemangioma of infancy and early childhood and the lymphangioma (both of which are unencapsulated) and the encapsulated cavernous hemangioma found typically in adults.
- pleomorphic adenoma of the lacrimal gland, dermoid cyst, neurilemmoma..are encapsulated.
- Non-Hodgkin lymphoma: can affect the entire orbit or can be confined to compartments of the orbit such as the lacrimal gland.
- Metastases to the orbit may present with distinctive signs and symptoms that point to the origin of the tumor. For example, metastatic prostatic carcinoma may present clinically like idiopathic orbital inflammation; metastatic neuroblastoma and Wilms tumor—richly vascular neoplasms—may produce characteristic periocular ecchymoses. Neoplasms may also invade from the sinuses into the orbit.
- In individuals with AIDS, Kaposi sarcoma may develop in either the eyelid or the conjunctiva. In the eyelid the lesion may appear clinically to have a purple hue because the vascular lesion is embedded in the dermis, but in the thin mucous membrane of the conjunctiva, Kaposi sarcoma appears bright red and may be confused clinically with a subconjunctival hemorrhage.

Conjunctiva

- palpebral conjunctiva, is tightly tethered to the tarsus and may respond to inflammation by being thrown into minute papillary folds as may occur in allergic conjunctivitis and bacterial conjunctivitis.

- Primary lymphoma of the conjunctiva (typically indolent marginal zone B-cell lymphoma) is most likely to develop in the fornix. The bulbar conjunctiva—the conjunctiva that covers the surface of the eye—is a nonkeratinizing stratified squamous epithelium.
- The conjunctiva, like the eyelid, is richly invested with lymphatic channels. Malignant neoplasms arising in the eyelid and conjunctiva tend to spread to regional lymph nodes (parotid and submandibular node groups).

CONJUNCTIVAL SCARRING

- infection with *Chlamydia trachomatis* (trachoma) may produce significant conjunctival scarring.
- may be scarred iatrogenically through reaction to drugs or as a consequence of surgery.
- KERATITIS AND ULCERS [cornea]
 - Various pathogens—bacterial, fungal, viral (especially herpes simplex and herpes zoster), and protozoal (*Acanthamoeba*)—can cause corneal ulceration.

CATARACT

- lenticular opacities that may be congenital or acquired
- Systemic diseases (such as galactosemia, diabetes mellitus, Wilson disease, and atopic dermatitis), drugs (especially corticosteroids), radiation, trauma, and many intraocular disorders are associated with cataract.
- Age-related cataract typically results from opacification of the lens nucleus (nuclear sclerosis).
- The technique that is most commonly used to remove opacified lenses extracts the lens contents, leaving the lens capsule intact (extracapsular cataract extraction).
- A prosthetic intra-ocular lens may be inserted into the eye.

THE ANTERIOR SEGMENT AND GLAUCOMA

- The term glaucoma refers to a collection of diseases characterized by distinctive changes in the visual field and in the cup of the optic nerve. Most of the glaucomas are associated with elevated intra-ocular pressure, although some individuals with normal intra-ocular pressure may develop characteristic optic nerve and visual field changes (normal or low-tension glaucoma).
- glaucoma can be classified into two major categories. In **open-angle glaucoma** the aqueous humor has complete physical access to the trabecular meshwork, and the elevation in intra-ocular pressure results from an increased resistance to aqueous outflow in the open angle. In **angle-closure glaucoma** the peripheral zone of the iris adheres to the trabecular meshwork and physically impedes the egress of aqueous humor from the eye.

Uvea:

- contain iris, the choroid and ciliary body
- **UVEITIS**
 - The term *uveitis* can be applied to any type of inflammation in one or more of the tissues that compose the uvea.
 - Uveitis may be caused by infectious agents (e.g., *Pneumocystis carinii*), may be idiopathic (e.g., sarcoidosis), or may be autoimmune in origin (sympathetic ophthalmia).

- Uveal melanoma is the most common primary intra-ocular malignancy of adults.
- The most common intra-ocular malignancy of adults is metastasis to the uvea, typically to the choroid.

Retina and Vitreous

- The retina therefore responds to injury by means of gliosis.

RETINAL DETACHMENT

- Rhegmatogenous retinal detachment is associated with a full-thickness retinal defect. Retinal tears may develop after the vitreous collapses structurally, and the posterior hyaloid exerts traction on points of abnormally strong adhesion to the retinal internal limiting membrane. Liquefied vitreous humor then seeps through the tear and gains access to the potential space between the neurosensory retina and the RPE. Re-attachment of the retina to the RPE generally requires relief of vitreous traction through indenting of the sclera by surgical procedures.
- Non-rhegmatogenous retinal detachment (retinal detachment without retinal break) may complicate retinal vascular disorders associated with significant exudation and any condition that damages the RPE and permits fluid to leak from the choroidal circulation under the retina. Retinal detachments associated with choroidal tumors and malignant hypertension are examples of non-rhegmatogenous retinal detachment.

Diabetes mellitus and retinopathy

- diabetic microangiopathy in general, the basement membrane of retinal blood vessels is thickened. In addition, the number of pericytes relative to endothelial cells diminishes. Microaneurysms are an important manifestation of diabetic microangiopathy.
- Structural changes in the retinal microcirculation have been associated with a physiologic breakdown in the blood-retinal barrier. Thus, the retinal microcirculation in diabetics may be exceptionally leaky, giving rise to macular edema, a common cause of visual loss in these patients.

RETINAL NEOPLASMS

Retinoblastoma

- the cell of origin of retinoblastoma is neuronal.
- retinoblastoma occurs in individuals who inherit a germline mutation of one RB allele. Retinoblastomas arising in the context of germline mutations not only may be bilateral.
- Tumors may contain both undifferentiated and differentiated elements. The former appear as collections of small, round cells with hyperchromatic nuclei. In well-differentiated tumors there are Flexner-Wintersteiner rosettes and fleurettes reflecting photoreceptor differentiation.

Retinal Lymphoma

- Primary retinal lymphoma is analogous to primary large-cell lymphoma of the brain; therefore, it involves the two retinal layers derived from brain: the neurosensory retina and the RPE. The underlying choroid is typically filled with a cytologically benign lymphoid infiltrate. Primary intra-ocular lymphoma tends to occur in older individuals and may mimic uveitis clinically. The diagnosis depends on a demonstration of lymphoma cells in vitreous aspirates.

EXIT EXAM NOTE ON CNS PHARMACOLOGY

General Anesthetics

- Anaesthesia is the reversible loss of response to a noxious stimuli. It may be general anaesthesia (if associated with loss of consciousness) or local anaesthesia (consciousness is maintained).
- Four main features of balanced anaesthesia are:
 - Unconsciousness, Muscle relaxation, Analgesia, Abolition of compensatory reflex responses.
- **Stages and planes of anesthesia:** to indicate depth of anesthesia.
 - a. Stage I: Analgesia, amnesia
 - b. Stage II: Loss of consciousness (may be excitement)
 - c. Stage III: Surgical anesthesia
 - (1) Four planes have been described relating to increased depth of anesthesia. Plane IV includes maximal pupil dilation, apnea, and circulatory depression.
 - (2) The loss of the eyelash reflex and a pattern of respiration that is regular in rate and depth are the most reliable signs of stage III anesthesia.
 - d. Stage IV: Respiratory and cardiovascular failure.
- **Inhalation anesthetics**
 - Include: Nitrous oxide [gas], isoflurane, desflurane, sevoflurane, halothane, and enflurane [the rest are volatile liquid].
 - **MOA:**
 - **GABA (A) -receptor chloride channels:** these anesthetic agents directly and indirectly facilitate a GABA-mediated increase in chloride (Cl^-) conductance to hyperpolarize.
 - **Ligand-gated potassium (K^+) channels:** these anesthetic agents increase potassium conductance to hyperpolarize and inhibit neuronal membrane activity.
 - **Pharmacokinetics**
 - Generally : anesthetics inhaled $\rightarrow$ alveolar tension $\rightarrow$ arterial tension $\rightarrow$ brain/ tissue tension $\rightarrow$ venous tension $\rightarrow$ back to alveolar tension $\rightarrow$ exhaled.
 - Factors affecting Kinetics
 - Pulmonary ventilation: Increased rate and depth of ventilation, will increase the partial pressure more rapidly, increasing on set of anesthetic action.
 - Blood gas partition coefficient [BGPC]: the greater the BGPC, the increased solubility in blood, so decreased free form, so take more time to

establish partial pressure in arterial blood , so lesser the speed of onset and recovery.

- Increased pulmonary flow, for example, increased cardiac output decreases the rate of rise in partial pressure by presenting a larger volume of blood into which the anesthetic can dissolve, so decrease the rate of anesthetics induction.
- The higher the lipid: blood partition coefficient, the faster the onset of action of anesthetics.
- Minimum alveolar concentration (MAC) is a relative term defined as the minimum alveolar concentration at steady state (measured in volume/volume percent) that results in immobility in 50% of patients when exposed to a noxious stimulus.
- This agents have cardiac and respiratory depressant effect.

Characteristics of selected general anesthetics

Inhaled anesthetics

▪ Nitrous Oxide ['laughing gas'.]

- colourless, non-irritating and non-inflammable, poor muscle relaxant.
- shows faster induction & recovery of anaesthesia (low blood gas partition coefficient).
- used in combination with other inhalation anesthetics to increase their rate of uptake and to add to their analgesic activity.
- it increases the volume of uptake of a second blood-soluble gas such as halothane (second-gas effect), which then speeds the induction of anesthesia.
- It is used as a supplement to anaesthesia (because it is not a complete anaesthetic).
- commonly associated with postoperative nausea and vomiting.
- inactivates methionine synthase , inhibition of DNA and protein synthesis and result in megaloblastic anemia or leukopenia.

▪ Halothane

- colourless, volatile liquid
- non-irritant, non explosive and pleasant smelling agent.
- good anaesthetic but very poor analgesic agent.
- inhalational agent of choice in bronchial asthma due to its bronchodilator action.
- cause hepatitis on repeated use.
- result in malignant hyperthermia, which can be treated with dantrolene.

Intravenous anesthetics

▪ Ketamine

- phencyclidine (hallucinogenic) derivative
- blocking NMDA receptors of glutamate.
- producing dissociative anaesthesia (state of profound analgesia, amnesia with light sleep, immobility, feeling of dissociation from one's own body and the surroundings).
- Hallucinations are the most common side effect.
- intravenous anaesthetic agent of choice for induction in children.

▪ Propofol

- It is a milky white powder.

- Rapid onset and short duration.
- strong antiemetic and antipruritic action.
- used for outpatient surgical procedures, in intensive care settings, and in balanced anesthesia.
- pain at injection site and systemic hypotension from decreased systemic vascular resistance.
- Potentiate chloride current mediated through GABA_A receptor.
- **Thiopentone**
 - ultra short acting barbiturate
 - rapid onset (20 seconds) and has a short duration of action [highly lipid soluble].
 - possesses anticonvulsant action
 - Instead of producing analgesic effect, it can produce hyperalgesia
 - contra-indicated in acute intermittent porphyria.
- **Others**
 - Benzodiazepines: not analgesic agents but possess muscle relaxing and anticonvulsant property. Lorazepam, Midazolam.
 - Droperidol : is a D₂ receptor blocker, used along with fentanyl to produce neurolept analgesia and neurolept anaesthesia.
 - Opioids: like Fentanyl, alfentanil. very strong analgesic activity.

Sedative Hypnotic Drugs

- Sedative is a drug that calms a person whereas hypnotics induce sleep.
- CNS depressant drugs can produce sedation, hypnosis, anaesthesia and coma depending on the dose.
- Major group of drugs useful for the treatment of insomnia include barbiturates, benzodiazepines and newer drugs.

Barbiturates

- act by increasing the Cl⁻ conductance across GABA_A-BZD-Cl⁻ channel complex.
- have GABA mimetic as well as GABA facilitatory action to increase the duration of Cl⁻ channel opening.
- indications of barbiturates in clinical practice are epilepsy (phenobarbitone) and anaesthesia (thiopentone as inducing agent).
- powerful enzyme inducing agents and are prone to several drug interactions.
- high abuse liability and may precipitate withdrawal symptoms.
- no specific antidote is available.
- Cause distortion of sleep architecture by decreasing the duration of REM.[result in hangover].
- Absolutely contraindicated in acute intermittent porphyria (because porphyrin synthesis is increased due to induction of d-ALA synthase; rate limiting enzyme in porphyrin synthesis, by barbiturates).

Benzodiazepines

- Bind to (BZD receptor) and increase the binding of GABA to GABA [A] receptor (GABA facilitatory action). These drugs increase the frequency of Cl⁻ channel opening.
- Also possess anxiolytic, anticonvulsant and skeletal muscle relaxant properties.
- Compared to barbiturates: less hangover and less distortion of sleep architecture, Abuse liability is less.
- Can be treated with specific antidote, flumazenil [competitive antagonist].
- Most of the BZDs are metabolized in the liver to produce active products (thus long duration of action, eg. Diazepam.
- Lorazepam, oxazepam,; directly conjugated without metabolism to active products. These drugs are thus short acting.
- Exhibit cross tolerance with other sedative–hypnotic agents, including alcohol and barbiturates.

Newer Hypnotic Drugs

- Zolpidem, zaleplon, and eszopiclone
 - used for short-term treatment of insomnia
 - act on a specific subtype of the benzodiazepine receptor BZ1 (omega1).
 - reduced potential compared to the benzodiazepines for tolerance, abuse, physical dependence, or rebound insomnia
- Buspirone
 - nonbenzodiazepine that selectively relieves anxiety without the sedation, hypnosis, general CNS depression, or drug abuse liability of the benzodiazepines.
 - used primarily to treat generalized anxiety disorder
 - partial agonist at serotonin (5-HT_{1A})-receptors.

Alcohol

- All alcohols cause CNS depression, in part through GABA mimetic activity.
- Chronic alcoholism:
 - 1. Hypoglycemia
 - 2. Fatty liver and lipemia
 - 3. Muscle wasting (long-term alcoholic, poor food intake)
 - 4. Gout (lactate competes with urate for excretion)
- Disulfiram inhibits Acetaldehyde Dehydrogenase (mitochondria) to treat alcohol dependence.

DRUGS USED IN PARKINSONS

- Etiology: low DA, high Ach. Thus the rationale for rx is to inc DA or dec Ach
- Impt': DA is an inhibitory NT. But since it inhibits GABAergic neurons, it leads to excitation (such as the starting of mov't...)
- The bradykinesia observed is due to low DA while the muscle rigidity & resting tremor are due to high Ach (The two classes of drugs act accordingly)

- 1) Drugs that increase DA levels:

Levodopa: must know!

- Prodrug that is converted to DA by B- decarboxylase. (we can't use DA itself since it can't cross the BBB)
- **Needs co-administration of carbidopa** to inhibit its premature peripheral conversion to dopamine. (Carbidopa doesn't cross the BBB so it doesn't prevent L-dopa's central conversion)
 - o L-dopa alone can have GI effects (stimulates vomiting in CTZ; better with/after meal) and low cardiac stimulatory effects (inc catecholamines). All these peripheral effects are reduced with the use of carbidopa
- Effective for only about 5 yrs & it shows fluctuations ("on-off phenomenon") so DA recep agonists are preferred as initial rx
- Proposed solns for fluctuations:
 - o dec dose and inc frequency of dosage, reduce protein intake (A.As compete with transport mechanism of L-dopa in gut & BBB), co-administration of DA agonists, taking drug holidays etc...
- Drug interactions : Pyridoxine (vit B6) facilitates peripheral conversion of L-dopa.
 - With MAOI – hypertensive crisis
- CI:
 - o Psychosis (Since it inc DA levels, it can lead to schizophrenia)
 - o Angle closure glaucoma (but can be used in open angle)
 - o Increase GI bleeding so not used in active PUD
 - o Increases melanin so not used in malignant melanoma
 - o Cardiac diseases

DA recep agonists:, Pramipexole, ropinirole

- Advantageous b/c no fluctuations, enzyme conversions, food interactions or toxic metabolites
- Other uses: acromegaly, hyperprolactinemia (ab lactation)
- CI in psychotics & recent MI
- Divided into 2:
 - o Ergot derivatives: **Bromocriptine**, Pergolide
 - o Non-ergot derivatives:
 - **Pramipexole** (selective to D3 recep)→ Has **impact on disease progression**; excreted unchanged in urine
 - Ropinirole(selective to D2 recep); metabolized in the liver

Tolcapone & Entacapone:

- COMT inhibitors. (COMT converts DA to its metabolite 3-O-methyl dopa w/c is a partial agonist. Thus, it can act as an antagonist in the presence of DA & interfere with its action)

Selegiline & Rasageline :

- MAO B – selective inhibitor. (so dec degradation of DA)
- No tyramine interaction since tyramine is metabolized by MAO A (← according to Kaplan. But the lecturer actually listed risk of hypertensive crisis as a major concern)
- **Selegiline claimed to affect disease progression**
- 2) Drugs that dec Ach (antimuscarinics)
 - They must cross BBB: Benztropine, Biperiden, Procyclidine, Trihexyphenidyl, **&Diphenhydramine , w/c is an anti-histamine.**
 - o N.B: All anti-histaminic drugs have anti-muscarinic actions as well
 - Improve tremor and rigidity; not bradykinesia
 - Inferior to those that inc DA b/c the pathophysiology of parkinsons itself is in the dec of DA
 - Atropine-like adverse effects

ANTI-EPILEPTICS

- Epilepsy: sudden, excessive depolarization of some or all cerebral neurons.
 - o Can be focal(involving a part of the cortex), generalized (simultaneously affecting all cortical neurons) or absence seizures (a brief period of cerebral depression due to depolarizations in the thalamus)
- Thus, the rationale for rx is:
 - o Dec axonal conduction by preventing Na⁺ influx through fast Na channels:
 - Carbamazepine, Phenytoin
 - o Inc inhibition by facilitating GABA
 - Barbiturates, Benzodiazepines, Gabapentine
 - o Dec excitatory effects of glutamate
 - Lamotrigine, Topiramate, Felbamate
 - o Dec presynaptic Ca influx through type T channels in thalamic neurons (**can be used for absent seizures unlike the others**)
 - Valproic acid, Ethosuximide
- **Phenytoin**
 - o Blocks inactivated Na channels
 - o Is 90% bound to plasma albumin so if displaced → toxic
 - o **Is an inducer of CYP 450**
 - o Has zero order kinetics (same amount of drug eliminated regardless of dose) so reach toxic levels quickly –low TI
 - o Side effect: **Gingival hyperplasia, hirsutism**
 - SE common to all anticonvulsants: Osteomalacia, Megaloblastic anemia, Anaplastic anemia
 - o Teratogenic: cleft lip/palate
- Carbamazepine
 - o DOC for trigeminal neuralgia, focal & secondary generalized epilepsy
 - o Induces CYP including its own metabolism (autoinduction)

- Side effects: Steven Johnsons (exfoliative dermatitis) although more common with sulfonamides, liver & kidney dysfunction
- Teratogenic: cleft lip/palate, spina bifida
- Valproic acid/ valproate:
 - Similar to phenytoin. But also inhibition of GABA transaminase & blockade of Ca channels
 - **The only anticonvulsant that INHIBITS cyp450 so taken with an OCP**
 - Side effect: Alopecia (as opposed to hirsutism of phenytoin), weight gain, ketone metabolites may cause confusion in urine testing in DM.
 - Teratogenic: spina bifida
- **Phenobarbital: safest during pregnancy** (b/c of its high plasma protein binding so doesn't cross much to the placenta)
- Benzodiazepines:
 - For status epilepticus: **Lorazepam** , Diazepam
- Barbiturates:
 - Phenobarbital & primidone (prodrug of phenobarb)
- Lamotrigine:
 - Block Na channels & glutamate recep. Also Ca channel blocker.
 - A/E : Steven Johnson syndrome

Condition	Drug of choice
Grand mal seizure (general)	Valproate , Lamotrigine
Petit mal (absence seiz)	Valproate, Ethosuximide
Partial seizures	Valproate, carbamazepine
Status epilepticus	Lorazepam, diazepam (IV); Alternatively (for established status): phenobarb, phenytoin; for refractory status: thiopental, propolol , midazolam with intensive care

- Rx should always start as monotherapy and never stopped suddenly (taper dose).
Choice of drug is also determined by patient's age & sex.

OPIOID ANALGESICS

Strong agonist	Morphine, Methadone, Meperidine
Mild to moderate agonist	Codeine, propoxyphene, diphenoxylate
Mixed agonist- antagonist (partial agonist)	Pentazocine, Nalbuphene

- Endogenous opioids: endorphins, enkephalins
- Act through 3 receptors: μ , κ , δ . All are **Gi coupled**, i.e, they inhibit Ca channels and activate K channels

- μ recep is mainly the one responsible for effects of opioids such as: analgesia, euphoria, respiratory depression & physical dependence
- **Effects:**
 - o Analgesia
 - Modulate pain by disinhibition (by inhibiting the inhibitory neuron GABA)
 - Used in chronic pain. Not much effective in the rx of sharp pain
 - **Exacerbates colic pain** since it increases the tone of the gall bladder. So must be combined with anticholinergics for a patient with gall bladder stones
 - o Euphoria, thus leads to addiction. Also dysphoria in rare instances
 - o CNS depression/ Sedation
 - o **Low** cardiac depressant effect (not much recep in the heart); can be used as an anesthetic in cardiovascular surgery
 - o Respiratory depression by inhibiting respiratory center in the brainstem
 - It can be used as adjunct in rx of pulmonary edema in order to improve the prognosis since it reduces anxiety, cardiac preload & afterload. (N.B 1st line rx for pulm edema are diuretics)
 - o Antitussives (cough suppression)
 - The relatively weaker ones like codeine used in small amount. N.B. Codeine is not used in Ethiopia/ banned
 - o Increases tone & reduces motility of GI (leads to constipations) so Loperamide used in rx of diarrhea
 - o Activate CTZ : Nausea & vomiting
 - o Exacerbates ICP
- o Opioid Antagonists:
 - **Naloxone is given IV as the antidote** for acute poisoning with opioids (not p.o b/c of FPE)
 - Naltrexone: used for addiction mgt b/c of long $t_{1/2}$

NSAIDS/ NON- OPIOD ANALGESICS

- All have antipyretic, analgesic & anti- inflammatory effects
 - Inhibit COX; dec PG synthesis (w/c is the main reason for the adverse effects)
 - o Pgs normally act through their metabotropic receps (GPCR)
 - COX 1 : constitutively expressed.
 - o Indomethacin & Sulindac mainly act through its inhibition
 - COX2 : inducible; brings about inflammation, pain & fever
 - o Celecoxib & Rofecoxib are selective inhibitors. Known to have high cardiac mortality.
 - Ibuprofen & Meclofenamate equally inhibit both COX 1 and COX2
- Adverse Effects:
- Has antiplatelet action. Thus bleeding. (caution in pts undergoing surgery)
 - When COX pathway is dec, the lipoxygenase path is inc. Thus exacerbation of asthma

- Are weak acids so can bring about GI irritation directly. Thus, advised to always take NSAIDS after food
- Compromise renal BF by vasoconstriction → kidney damage
- **Asprin is CI for the use in viral infection due to the risk of developing Reye's Syndrome**
- When NSAIDs are taken with basic drugs, they have compromised absorption

Salicylates (eg. Asprin):

- Irreversible inhibitor of both COX 1 & 2
- Used in CV morbidities (antiplatelet action) & said to prevent colon cancer
- Effect on respiratory system
 - o Therapeutic dose: increases alveolar ventilation
 - o Higher doses: hyperventilation & respiratory alkalosis w/c is manageable by kidney buffer system
 - o Toxic dose: causes central respiratory depression
- Since it dec protective PG in the GI (PGI, PGE, PGF) → PUD. **Mesoprostol administration is protective**
- Reduces PGE & PGI in the kidney → salt & water retention, hyperkalemia, interstitial nephritis → "analgesic nephropathy"
- Pseudo- allergy risk due to inc leukotrienes

Indomethacin:

- More potent than asprin (since has more anti-inflammatory effects)
- 1st line rx for gout. **Asp is CI in gouty arthritis** since asp is known to inc uric acid levels
- 100% cross-reactive with asp
- **Used in closure of PDA** (Patent Ductus Arteriosus)
- Affects antihypertensive effect of diuretics, B-blockers, and ACEI

Ibuprofen:

- Reversible inhibitor of COX w/c is inferior to Indomethacin
- For chronic rx of rheumatoid & osteoarthritis

Diclofenac:

- Accumulates in joint spaces so preferred for Rheumatoid Arthritis
- Low bioavailability due to FPE
- Metabolized by CYP 3A4; clearance not affected by renal failure

Acetaminophen/ Paracetamol

- **NOT an NSAID** since it is anti-pyretic, analgesic but **not anti-inflammatory**.
- Has minimal peripheral COX inhibitory action (the PGs in the GI, kidney.. are less affected so better adverse-effect profile)
- **Preferred for viral infection in children** (not associated with Reye's syndrome)
- Metabolized in the liver to a toxic metabolite. **Antidote: NAC** (N-acetyl Cysteine)

Disease modifying anti-rheumatic agents (not analgesics themselves, just supplements):

- **Methotrexate**, gold salts, chloroquine & hydroxychloroquine, D-penicillamine

DRUG DEPENDENCE AND ADDICTION

- Drug abuse: using a drug for non-medical purposes
- Drug misuse: using an incorrect route of administration, dosage etc...
- Addiction: **a brain disease** since drugs bring about adaptive changes in the brain that lead to a different state of being. (they hijack our brain)
- Through time, there will be a need to escalate the dose to maintain effect.
- Tolerance → Dependence → Addiction → Compulsion ("wanting without liking")
- Important signs of drug abuse

Opiates (morphine, heroine)	Myosis , Flushing (histamine), Coma
Cannabinoids (marijuana)	Conjunctivitis
CNS depressants (Barbiturates, Benzodiazepines, Ethanol)	Mydriasis , weak & rapid pulse

- Basically, withdrawal symptoms are the opposite of the action of the drug, i.e, if the drug is a CNS stimulant, then upon withdrawal the patient shows signs of depression.
- Treating drug intoxications:

Benzodiazepines	Flumazenil - antidote
Barbiturates & alcohol	No antidote
Opioids	Naloxone - antidote
For the CNS stimulants and hallucinogens (Amphetamines, Cocaine, Marijuana, PCP...)	Lorazepam (anticonvulsant) for the agitation. Add haloperidol (antipsychotics) if risk of psychosis presents

ANTIDEPRESSANT DRUGS

- depression
 - is an affective disorder characterized by changes in mood
 - caused by a *functional* deficiency of *monoamine transmitters* (norepinephrine "NE" and/or "5-HT") at certain sites in the brain.

ANTIDEPRESSANT DRUGS

- MAOIs
 - Phenelzine, tranylcypromine, and isocarboxazid -> irreversible and non selective
 - Moclobemide -> reversible and selective inhibitor of MAO-A
 - Deprenyl and selegine -> selective inhibitor of MAO-B
 - MAOIs have lethal food drug interactions

- Tyramine containing foods [cheese, wine, tap beer...] cause hypertensive crisis and intracranial bleeding
- TCAs [block other receptors]
 - Amitriptyline, clomipramine, imipramine
 - Blockage of α -adrenergic receptors → Postural hypotension
 - Blockage of Histamine H₁ receptors → Sedation, thus they cause weight gain
 - Blockage of Muscarinic receptors → dry mouth, urine retention, constipation, blurred vision
- Heterocyclic antidepressants
 - Amoxapine, bupropion, mirtazapine
- 5HT-NE reuptake inhibitors
 - Duloxetine, venlafaxine
- 5HT antagonists
 - Nefazodone, trazodone
- SSRI
 - Escitalopram, fluoxetine, fluvoxamine, paroxetine, sertraline
 - SSRIs have no autonomic side effects or food interactions
 - Cause anxiety and sexual dysfunction
 - For sexual dysfunction change the medication or give sildenafil

USE OF ANTIDEPRESSANTS

- OCD → SSRI [first line of therapy], clomipramine
- Enuresis → TCAs
- Chronic pain- TCA, venlafaxine, duloxetine
- Bulimia → fluoxetine
- ADHD → TCAs, atomoxetine

DRUG-DRUG INTERACTIONS:

- "Serotonin Syndrome": A problem associated with increase in 5-HT level.
 - e.g. MAOIs + SSRIs or SSRIs + lithium.
- hyperthermia, cognitive reactions, muscle rigidity, rapid changes in mental status and vital signs.

ANTIMANICS

- mania → too much NE and serotonin
- drugs for bipolar disorder
 - classic drug [lithium] and newer agents [carbamazepine, clonazepam, olanzapine, and valproic acid]
- lithium
 - narrow TI

- major side effects are polydipsia, polyuria, polyphagia, tremor and decrease thyroid function
- don't give lithium with thiazide diuretics or non steroidal antiinflammatory drugs as they decrease the clearance of lithium

CNS STIMULANTS

- Psychomotor stimulants [cause excitements, euphoria, decreased feeling of fatigue, and increase motor activity]
 - Methylxanthines(caffeine, theobromine, theophylline), nicotine, cocaine, amphetamine, atomoxetine, modafinil, methylphenidate.
 - Methylxanthines
 - increase calcium levels -> stimulation of heart
 - stimulates CNS via A2 receptors
- Psychotomimetic drugs
 - profound changes in thought pattern and mood
 - Lysergic acid diethylamide (LSD) [5HT2 receptor agonist], Phencyclidine (PCP),
 - Tetrahydrocannabinol (THC), Rimonabant [*inverse agonist of CB1 receptor that was used for smoking cessation and facilitate weight loss*]
- Convulsants and respiratory stimulants
 - Doxapram and amiphenzole [used in acute respiratory failure], strychnine, bicuculline, picrotoxin, pentylenetetrazol (PTZ)
- Cocaine and amphetamines
 - ↑ DA related to reward phenomenon or drug seeking
 - *Amphetamine induces release of NTs while cocaine stops the reuptake*
 - Amphetamines suppress appetite unlike cannabinoids [marijuana]
 - Amphetamines -> used to treat incontinence, obesity, ADHD, Narcolepsy
 - CNS EFFECT AFTER USE OF amphetamines->performance of simple mental tasks is improved, however although more work is accomplished, the number of errors increases;If athletes are found using it-> disqualification
 - Cocaine-> used as a local anesthetic
 - Caffeine is used to treat apnea in new born when given IV

TREATMENT

- For withdrawal effects: Diazepam, clonidine
- For acute symptoms: IV diazepam, propranolol, verapamil
- Anti-psychotics & anti-depressants
- Hallucinogens
 - LSD [agonist of 5HT2 receptors]
 - Aversive not reinforcing [doesn't cause addiction]
 - synesthesia, perception of sensory modalities crosses over; sounds can be seen and sights can be heard
- Marijuana is the plant while hashish is the resin extracted from the plant

- THC is a chemical in marijuana that binds to CB receptors
- Binding to CB receptors leads to halting of GABA release
- Addiction-Compulsive drug seeking and use, even in the face of appalling consequences.
 - All addictive drugs activate the mesolimbic dopamine system and so an increase in dopamine leads to a pleasure state

OVERDOSE EFFECT

- Dilated pupils are seen in case of barbituates, benzodiazepines and ethanol overdose
- Constricted pupils are seen in case of heroin and opioid overdose
- No medical use as they cause high addiction potential-> flunitrazepam, heroin, LSD, mescaline, PCP, MDA, STP
- Medical used drugs that have the lowest abuse potential-> benzodiazepines, chloral hydrate, mild stimulants, most hypnotics weak opioids

EMERGENCY TREATMENT OF DRUG INTOXICATION

- Make sure to support vital functions in all drug intoxication
 - Alcohol-> none
 - Amphetamine-> lorazepam and haloperidol
 - Barbituates-> none
 - Benzodiazepines-> flumazenil
 - Cocaine-> lorazepam
 - Hallucinogen-> lorazepam
 - Opioids-> naloxone
 - Phencyclidine-> lorazepam and haloperidol

ANTIPSYCHOTIC DRUGS

- Schizophrenic patients show
 - Positive symptoms-> delusion, hallucination
 - Negative symptoms-> social withdrawal, lack of pleasure
 - Depression, cognitive defects [cant pay attention], catonia
- Psychosis producing drugs; levodopa, cocaine, amphetamine, khat, cathinone, apomorphine, phencyclidine [all addictive drugs increase the amount of dopamine]
 - Increase dopamine levels is correlated with schizophrenia
 - Studies show that schizophrenic patients have increased dopamine receptors
- Possible causes of schizophrenia-> excessive dopamine, stimulation of 5 HT_{2A/2C}, hypo-function of NMDA receptor [found in GABAergic inhibitory neurons]
- Partial DA agonists-> aripiprazole and bifeprunox used in schizophrenia
- Antipsychotic drugs could be used as antiemetic by acting in the CTZ [**D₂ RECEPTOR BLOCKING ACTION**]

IMPORTANT PATHWAYS AND SCHIZOPHRENIA

- Mesocortical pathway

- Excessive serotonin stimulation in this pathway is shown to decrease DA release; thus, leading to negative symptoms
 - Hence, atypical antipsychotics [potent blockers of serotonin receptors] are mainly used to treat negative symptoms
- Mesolimbic pathway
 - Excessive dopamine receptor stimulation leads to the positive symptoms
 - Typical and atypical antipsychotics block D2

ANTIPSYCHOTIC DRUGS

1. Older-agents/ typical/ conventional / **dopamine receptor antagonists**/ major tranquilizers/ neuroleptics
 - *Chlorpromazine, haloperidol, fluphenazine, thiridazine, loxapine, perphenazine*
2. Newer agents / atypical/ **serotonin-dopamine antagonists**/ 2nd generation/ **limbic specific**
 - *Clozapine, risperidone, olanzapine, quetiapine, ziprasidone, aripiprazole, asenapine, bifenox, paliperidone*

D2 RECEPTOR ANTAGONIST SIDE EFFECTS

- Effect on nigrostriatal pathway
 - Excess D2 block-> Parkinson's like effect, EPS, and could progress into an irreversible stage called tardive dyskinesia
- Effect on mesocortical pathway
 - Low dopamine exacerbates the negative symptoms
- Effect on mesolimbic path
 - Low dopamine improves the positive symptoms
- Effect on tuberoinfundibular path
 - Increase prolactin, infertility, sexual dysfunction
 - Majorly caused by the typical antipsychotic drugs due to their potent D2 receptor antagonism

TYPICAL Vs. ATYPICAL AGENTS

- Typical antipsychotic drugs have more EPS than atypical ones due to their potent D2 inhibitory effect on the nigrostriatal pathway
- Both treat positive symptoms well
- Atypical agents are more suitable to treat negative symptoms due to their serotonin receptor antagonist action
- Typical antipsychotic drug block a lot of extrareceptors other than D2 like;
 - Muscarinic, alpha-adrenergic, histamine receptor, serotonin

ADVERSE EFFECTS

- Muscarinic blockers
 - Loss of accommodation, urinary incontinence
- Adrenergic blocking
 - Orthostatic hypotension, impotence

- tardive dyskinesia
 - dopamine receptor supersensitivity after the receptors after a prolonged antagonism
- weight gain and D.M precipitation
 - mostly seen with clozapine and olanzapine
- acute dystonia, parkinsonism and akathisia occur early in treatment
 - acute dystonia-> sudden onset of brief or sustained abnormal posture, tongue protrusion
 - treat with benztropine, diphenhydramine or lorazepam
 - akathisia-> motor restlessness; '*ants in the pants*'
 - give beta blockers, anticholinergics, benzodiazepines
 - parkinsonism; give antiparkinsons drugs
- Tardive dyskinesia, tardive dystonia and tardive akathisia have a late onset /after years/ they are irreversible
 - Give Benzodiazepines, valproic acid, propranolol, clonidine
- Catalepsy symptoms
- Visual impairment: retinal deposits with thioridazine [causes browning of vision]
- Clozapine: agranulocytosis and seizures..
- Life threatening ventricular arrhythmias [ziprasidone and thioridazines]
 - Give beta blocker [atenolol]
- Neuroleptic Malignant Syndrome:NMS
 - a rare but serious side effect of neuroleptic (antipsychotic) that can be lethal.
 - Hyperthermia, Muscle rigidity Change in mental status Autonomic change e.g. sweating Tachycardia
 - give-amanatadine ,bromocriptine, dantrolene
- Clozapine
 - Best to treat
 - positive and negative symptoms, relapse
 - Most anticholinergic action
 - Most hypotensive
 - Has no EPS

ANTI ANXIETY DRUGS

- We don't need to treat mild anxiety; but, we should treat moderate anxiety, severe anxiety, panic, GAD, panic disorder, OCD, phobias and somatoform disorders
- Drugs used
 - First choice to treat anxiety = benzodiazepines
 - Alprazolam is primarily used to treat agoraphobia, panic attacks
 - Others-> oxazepam, diazepam, lorazepam
 - Since oxazepam doesn't need liver metabolism, it is used in elderly patients
 - 5HT agonist = buspirone
 - No withdrawal signs
 - Minimal abuse liability

- Slow onset of action and so it is not used for panic disorders
- Uses liver as its clearance
- Drug interaction
 - Rifampicin= inducer
 - Erythromycin= inhibitor
- Beta blockers
 - Used to treat somatic symptoms of anxiety
 - Decrease in blood pressure and heart rate
- Anti-depressants
 - First line agents in mgt. of chronic GAD
 - All anti-depressants have the same degree of anxiety reduction
 - SSRI are preferred because they have less side effect than TCAs
 - Doxepin, imipramine and desipramine are TCAs
 - Remember that TCAs have a wide variety of effects on other receptors
 - Atropine like action
 - Alpha blocking action
 - 5HT reuptake inhibitors
 - Fluoxetine
 - Used as a first line therapy to treat OCD
 - MAO= last choice due to their lethal food drug interaction
 - Phenelzine
 - Don't take with wine, beer, fermented foods and cheese
 - Milk is ok